



集成电路设计工程 2025秋

Lecture 9 版图设计1

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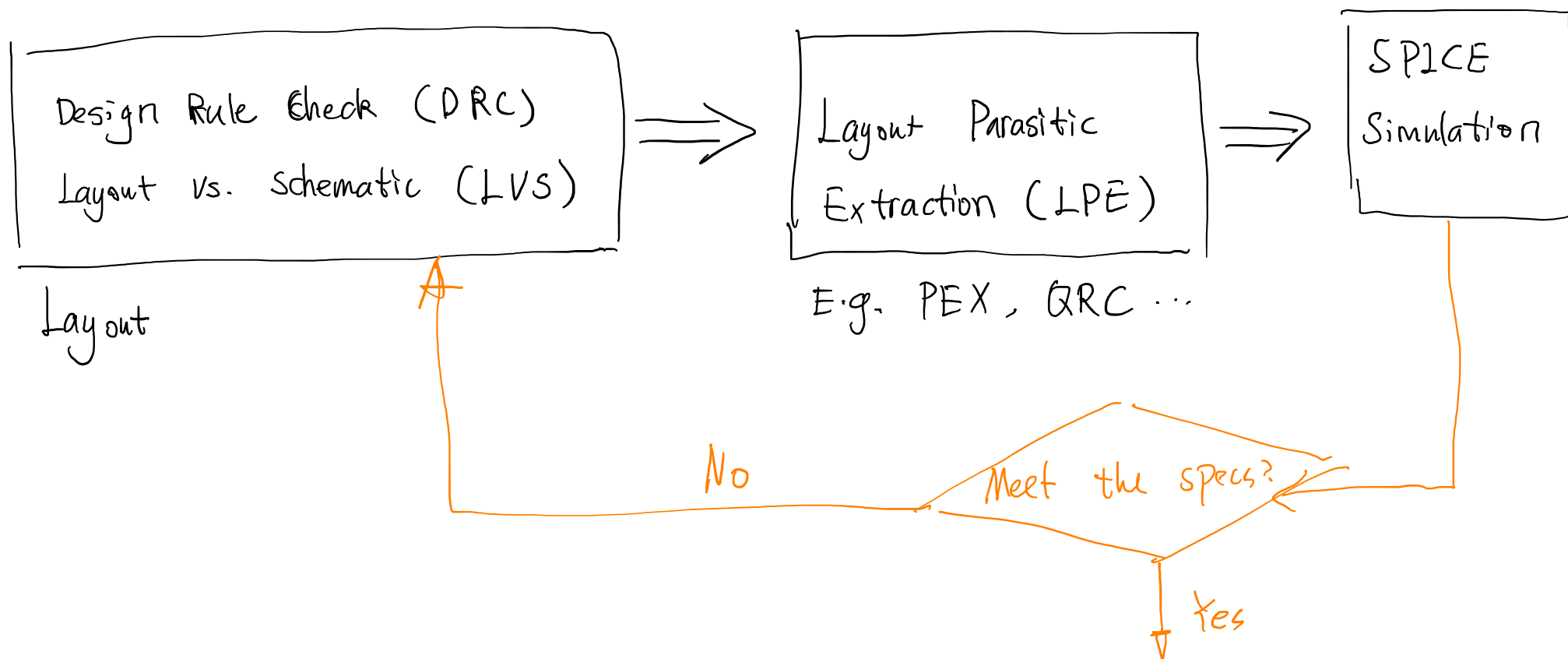


- Layout Design Flow and Layout Floorplan(版图布局)
- Layout of Building Blocks
- Layout Design and Bindkeys
- DRC/LVS/PEX
- Advanced Design Techniques in Chip Core Layout/Simulation

Layout Basics



Layout Iteration Flow

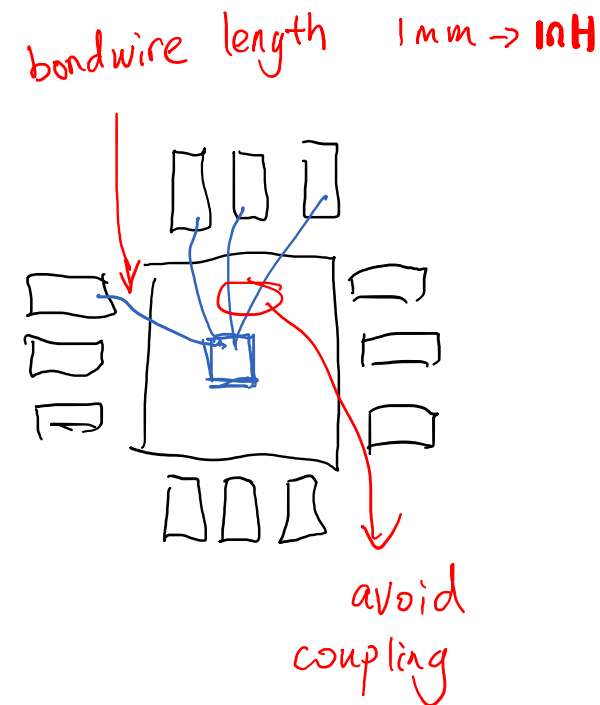


Layout Basics



Top-Level Floorplanning:

1. Estimate the area
2. choose package (e.g. QFN, COB, flip chip ...)
3. Arrange blocks and pads.



Floorplanning for a Single Block

① Place the component (Use ADEXL automatic placement as a starting point)

- Matching. (size $\uparrow$, $V_{GS} - V_{th} \uparrow$, be symmetric, add dummy devices ...)
- Signal flow
- Space for routing, substrate & well contacts.
 $n+$, $p+$

Layout Basics



② Plan pin locations

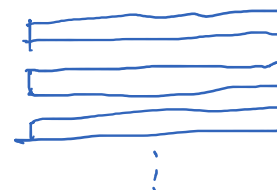
- Signal flow and connections to other blocks.

③ Plan power routing

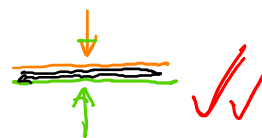
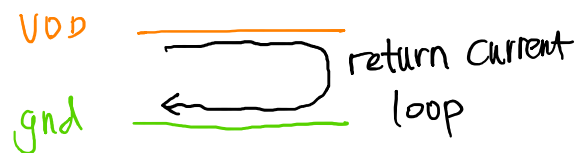
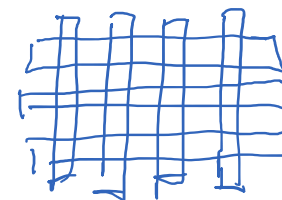
- IR drop $\Rightarrow$ use wide metal
- Return current $\Rightarrow$ keep vdd and gnd close.

width $< \sim 10\mu\text{m}$ (DRC)

parallel traces



Mesh

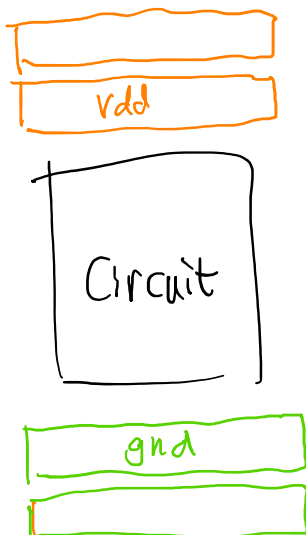


- bypass vdd to gnd (insert enough bypass capacitors between vdd & gnd)

Layout Basics

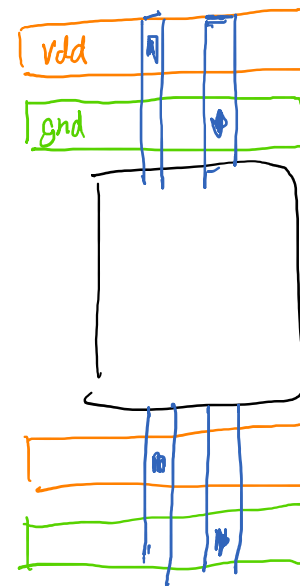


Example



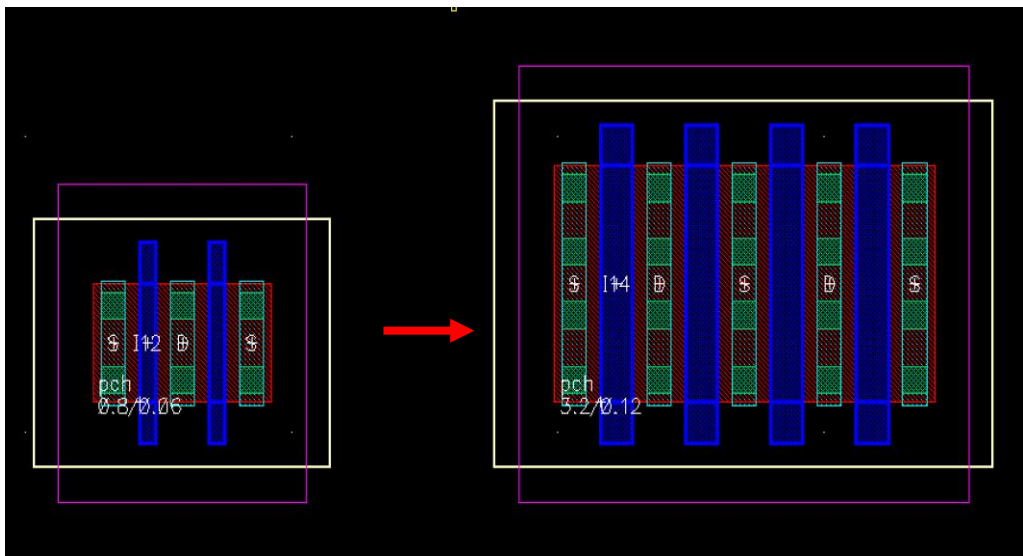
X bad layout

— Separate vdd & gnd of noisy blocks
clock, digital ...



✓ good layout
from quiet blocks.
↳ input/output, bias ...

1.1、版图器件介绍PMOS (tsmcN65)

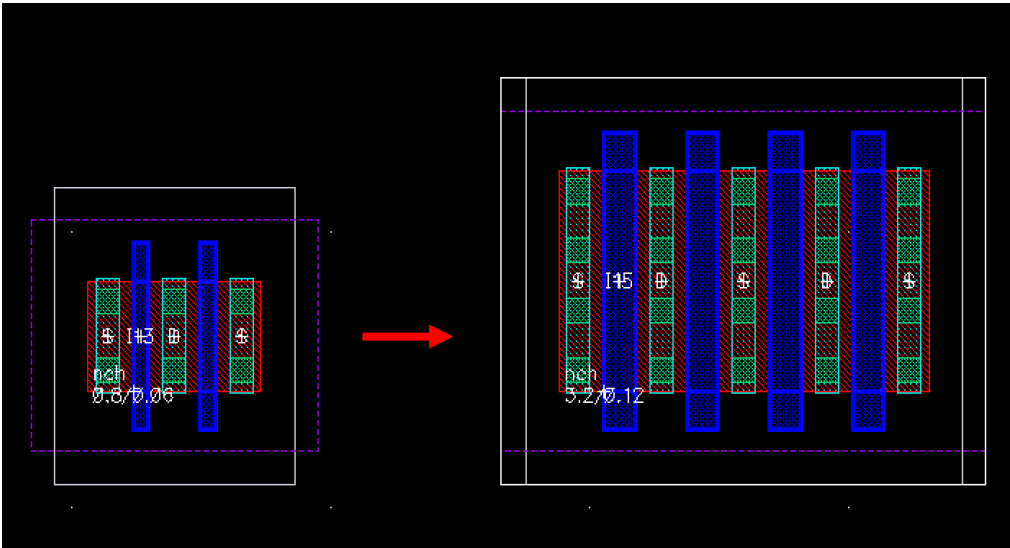


上图为PMOS版图，当版图尺寸由
L=60n、W=400n、 Fingers=2；改
为L=120n、W=800n、 Fingers=4；
版图的变化亦如上图所示；
右图为版图的参数设置；

Attribute	Connectivity	Parameter	Property	ROD
Model name		pch		
description		1.2V standard VT PMOS transistor		
I (M)		60n		
w (M)		400n		
total_width(M)		800n		
Number of Fingers		2		
total_m		2		
Hard_constrain		☒		
Calc Diff Params		☒		
Source_area		1.4e-13		
Drain_area		8e-14		
Source_periphery_(M)		2.3u		
Drain_periphery_(M)		1.2u		
NRS		0.125		
NRD		0.125		
Calc SA SB SD		☒		
fingers_SP_INC(M)		0		
SD(Fingers_Spacing)_(M)		200n		
SA(LOD_effect)_(M)		175.00n		
SB(LOD_effect)_(M)		175.00n		

Attribute	Connectivity	Parameter	Property	ROD
Model name		pch		
description		1.2V standard VT PMOS transistor		
I (M)		120.0n		
w (M)		800n		
total_width(M)		3.2u		
Number of Fingers		4		
total_m		4		
Hard_constrain		☒		
Calc Diff Params		☒		
Source_area		4.4e-13		
Drain_area		3.2e-13		
Source_periphery_(M)		5.9u		
Drain_periphery_(M)		4u		
NRS		0.03125		
NRD		0.03125		
Calc SA SB SD		☒		
fingers_SP_INC(M)		0		
SD(Fingers_Spacing)_(M)		200n		
SA(LOD_effect)_(M)		175.00n		
SB(LOD_effect)_(M)		175.00n		

1.2、版图器件介绍NMOS (tsmcN65)

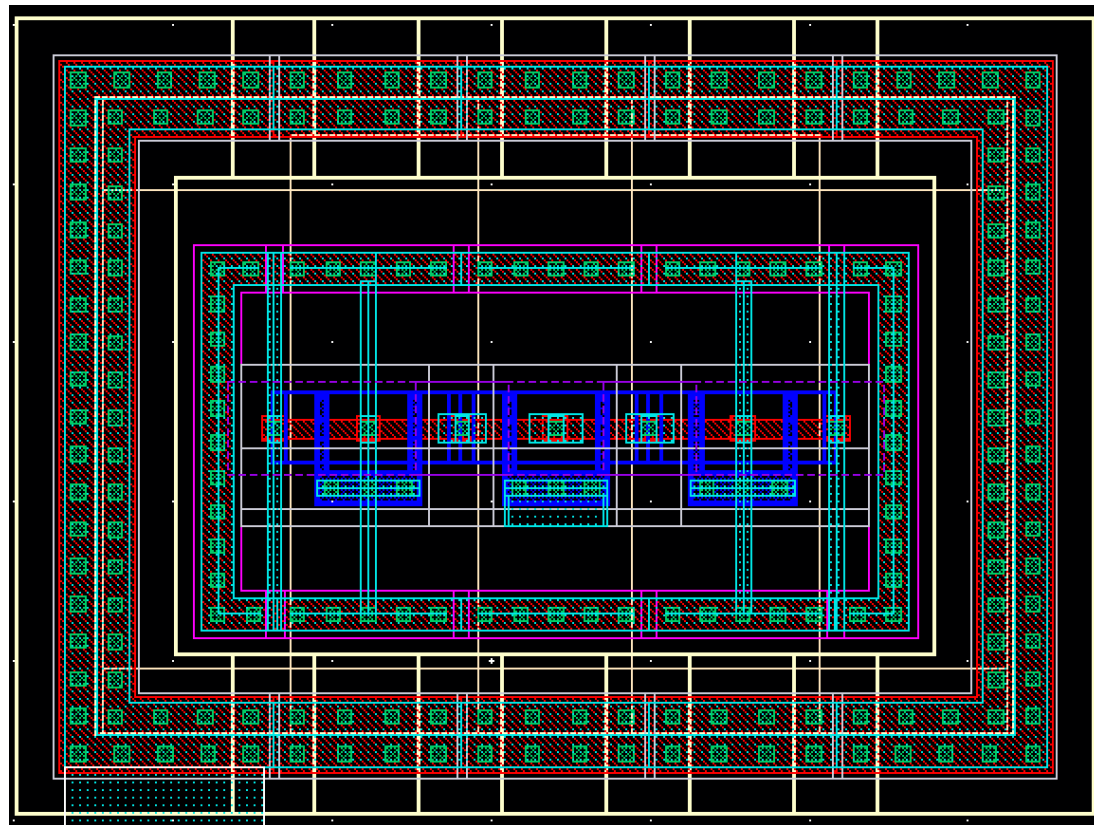


上图为NMOS版图，当版图尺寸由
L=60n、W=400n、Fingers=2；改为
L=120n、W=800n、Fingers=4；版
图的变化亦如上图所示；
右图为版图的参数设置；

Attribute	Connectivity	Parameter	Property	ROD
Model name		nch		
description		1.2V standard VT NMOS transistor		
I (M)		60n		
w (M)		400n		
total_width(M)		800n		
Number of Fingers		2		
total_m		2		
Hard_constrain		☒		
Calc Diff Params		☒		
Source_area		1.4e-13		
Drain_area		8e-14		
Source_periphery_(M)		2.3u		
Drain_periphery_(M)		1.2u		
NRS		0.125		
NRD		0.125		
Calc SA SB SD		☒		
fingers_SP_INC(M)		0		
SD(Fingers_Spacing)_(M)		200n		
SA(LOD_effect)_(M)		175.00n		
SB(LOD_effect)_(M)		175.00n		

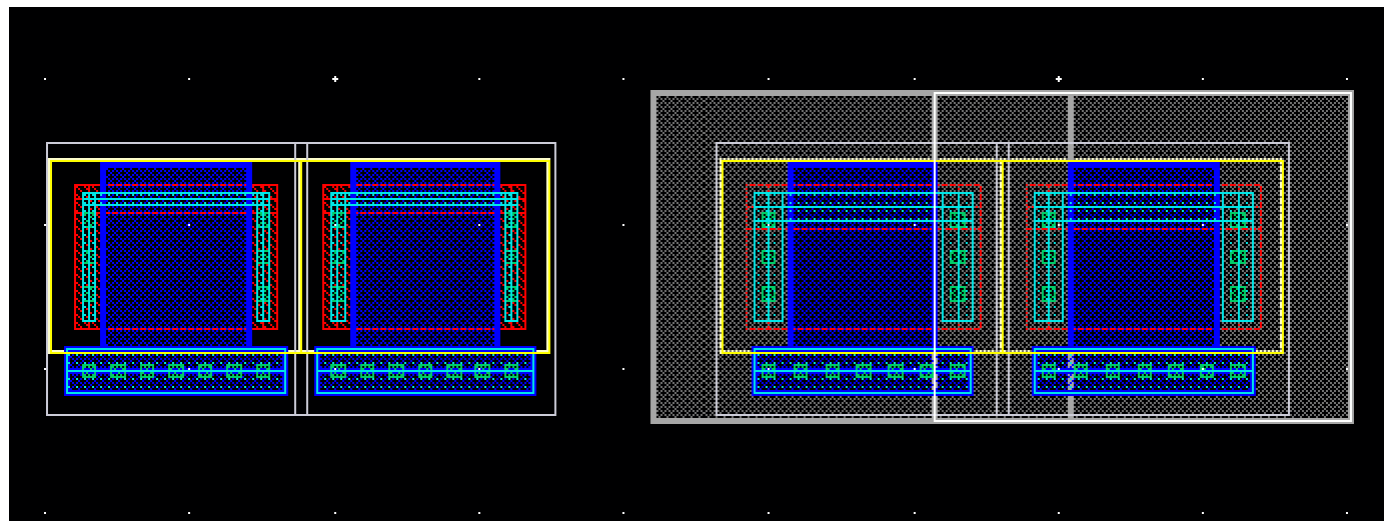
Attribute	Connectivity	Parameter	Property	ROD
Model name		nch		
description		1.2V standard VT NMOS transistor		
I (M)		120.0n		
w (M)		800n		
total_width(M)		3.2u		
Number of Fingers		4		
total_m		4		
Hard_constrain		☒		
Calc Diff Params		☒		
Source_area		4.4e-13		
Drain_area		3.2e-13		
Source_periphery_(M)		5.9u		
Drain_periphery_(M)		4u		
NRS		0.03125		
NRD		0.03125		
Calc SA SB SD		☒		
fingers_SP_INC(M)		0		
SD(Fingers_Spacing)_(M)		200n		
SA(LOD_effect)_(M)		175.00n		
SB(LOD_effect)_(M)		175.00n		

1.3、版图器件介绍DNW_NMOS (tsmcN65)



上图展示了DNW_NMOS的版图绘制方法；
右图为DNW_NMOS版图的参数设置；

Attribute	Connectivity	Parameter	Property	ROD
Model name		nch_lvt_mac		
description		1.2V LOW VT NMOS in DNW transistor		
I (M)		60n		
w (M)		120.0n		
total_width(M)		720.0n		
Number of Fingers		6		
total_m		6		
Hard_constraint		☒		
Mismatch_Sigma		1		
Calc Diff Params		☒		
Source_area		1.62e-13		
Drain_area		1.395e-13		
Source_periphery_(M)		3.6u		
Drain_periphery_(M)		3u		
NRS		0.243056		
NRD		0.243056		
Calc SA SB SD		☒		
fingers_SP_INC(M)		0		
SD(Fingers_Spacing_(M)		350.0n		
SA(LOD_effect_(M)		100n		
SB(LOD_effect_(M)		100n		



①nmoscap

②nmoscap25

上图展示了nmoscap和nmoscap25的版图绘制方法，二者的主要区别除了容值不同，nmoscap25的漏电比nmoscap要小，但二者都是广泛使用在集成电路中的mos电容；

右图为nmoscap 的参数设置；

Attribute	Connectivity	Parameter	Property	ROD
Model name		nmoscap		
CapValue@1.0V_(F)		17.33818f		
CapValue@0V_(F)		10.84023f		
CapValue@-1.0V_(F)		2.892874f		
description		1.0V or 1.2V Standard-Vt NMOS in N-Well varactor		
Length_(M)		1u		
Width_(M)		1u		
Hard_constrain		☒		

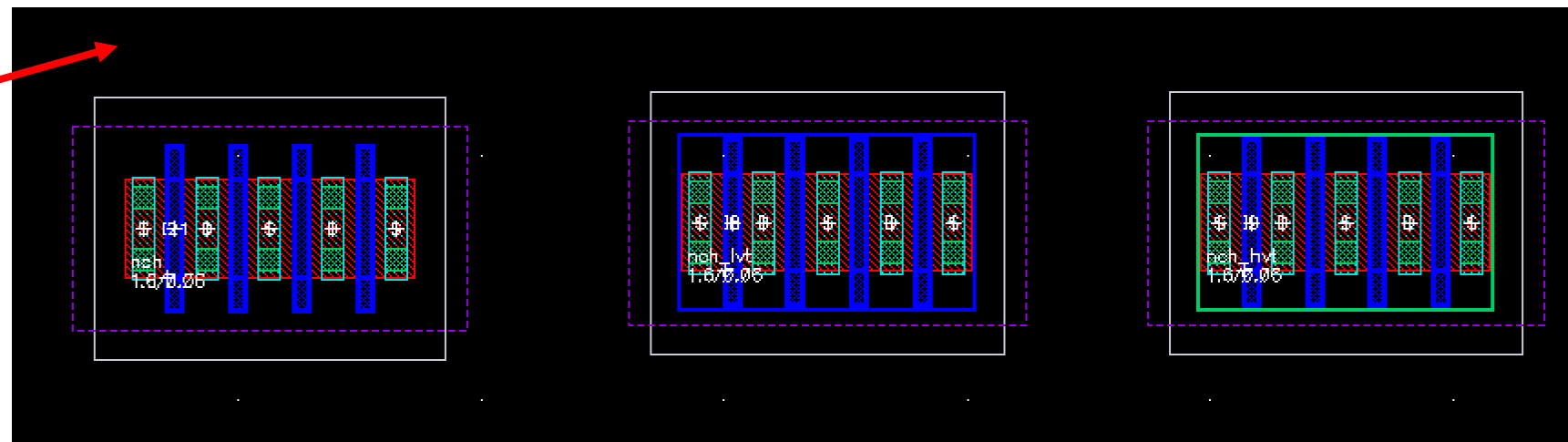
Attribute	Connectivity	Parameter	Property	ROD
Model name		nmoscap_25		
CapValue@2.5V_(F)		6.3444f		
CapValue@0V_(F)		4.8135f		
CapValue@-2.5V_(F)		1.9463f		
description		2.5V Standard-Vt NMOS in N-Well varactor		
Length_(M)		1u		
Width_(M)		1u		
Hard_constrain		☒		



1.5、版图器件介绍svt、lvt、hvt (tsmcN65)

右图展示了nch、nch_lvt、nch_hvt三种类型NMOS Core device，在版图上它们只有一层不同，而这一层即可以区分出它们的不同类型；

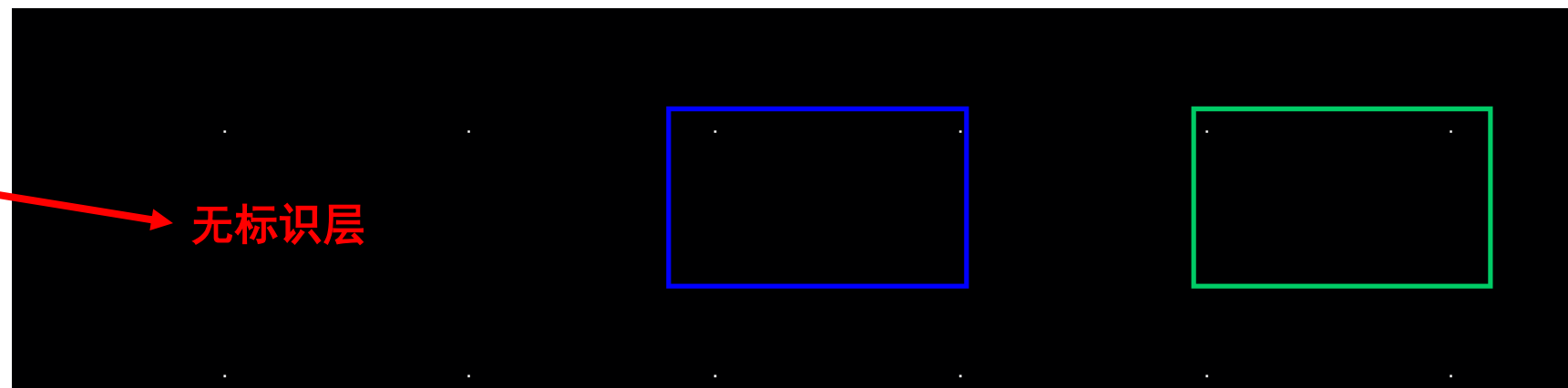
在tsmcN65工艺中，作为SVT类型的nch NMOS管以不添加任何标识层来判别出其SVT的性质；



①nch

②nch_lvt

③nch_hvt

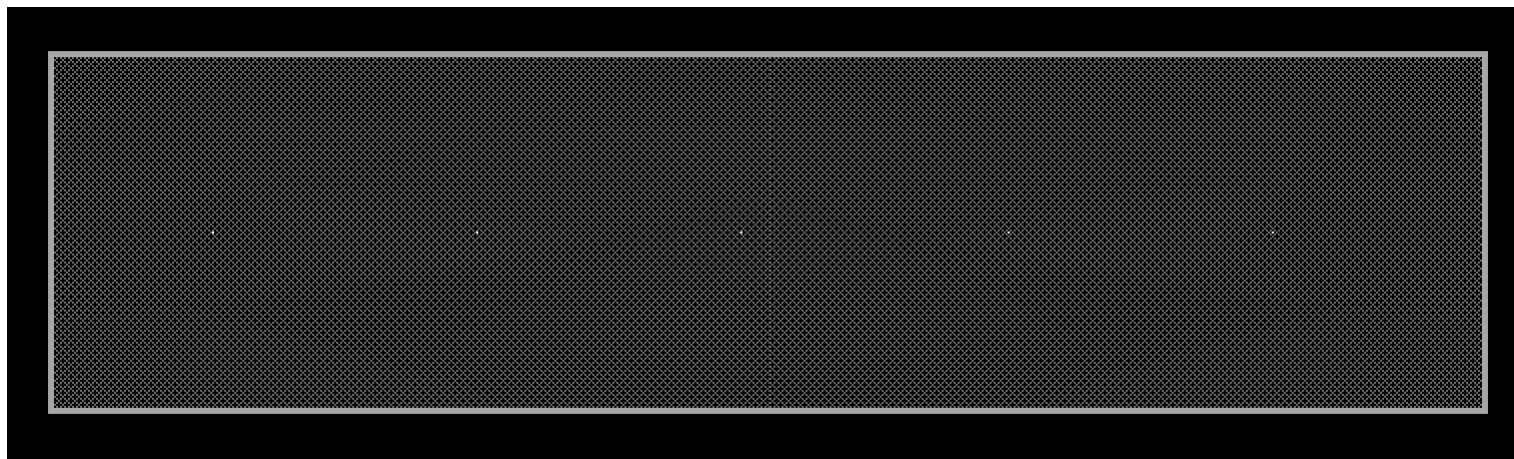
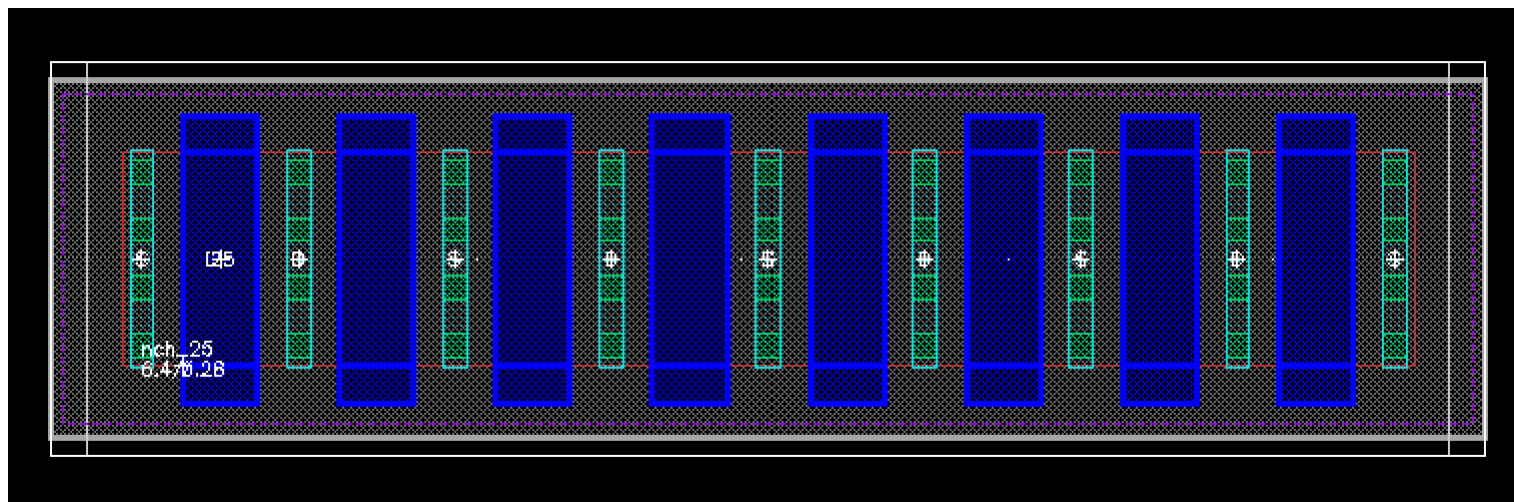


无标识层

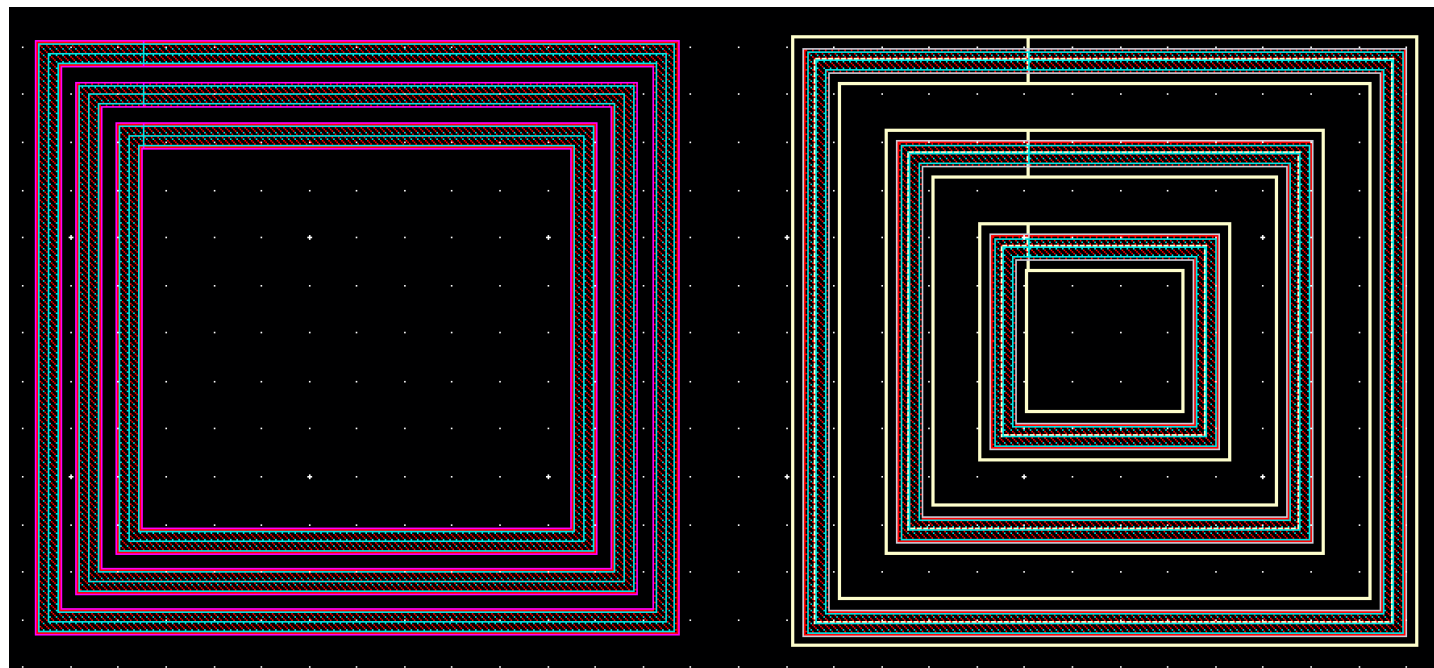
1.6、版图器件介绍2.5V (tsmcN65)



右图展示了nch_25类型的NMOS IO device, 同样, 与svt、lvt、hvt三种类型的NMOS Core device 相比, IO device用高压标识层OD_25进行区分;

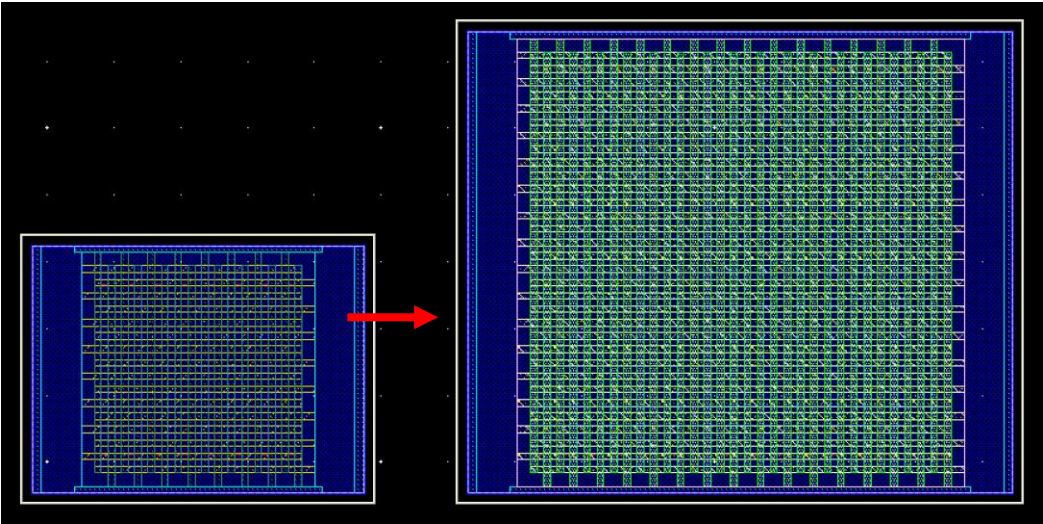


1.7、版图器件介绍GuardRing (tsmcN65)



上图为两种类型的GuardRing，分为Pwell（左）和Nwell（右），GuardRing的主要作用是作为N管和P管的衬底供电，以及对重要的版图模块进行保护等作用，也可以隔离噪声；

1.8、版图器件介绍MOM电容 (tsmcN65)



上图为MOM电容版图，当版图尺寸由W=100n、S=100n、h fingers=16、v fingers=16、RTMOM **BtoT**=M1-M3；改为W=100n、S=100n、h fingers=32、v finger=32、RTMOM **BtoT** =M1-M7；版图的变化亦如上图所示；

Attribute	Connectivity	Parameter	Property	ROD
Model name		crtmom		
description		CRTMOM		
fingers Width(M)		100n		
fingers Space(M)		100n		
Number of horizontal fingers		16		
Number of vertical fingers		16		
RTMOM Bottom Metal Layer		1		
RTMOM Top Metal Layer		3		
CapValue@0V_(F)		15.2599f		
Create_OD_dummy		☒		
OD_width(M)		100n		
Create_OD2_Layer		☐		
DFM_rule		☐		
WellType		N		
Adding_DMEXCL_Layer		☒		
Hard_constrain		☒		
Register_number		Tsinghua-2214524229		



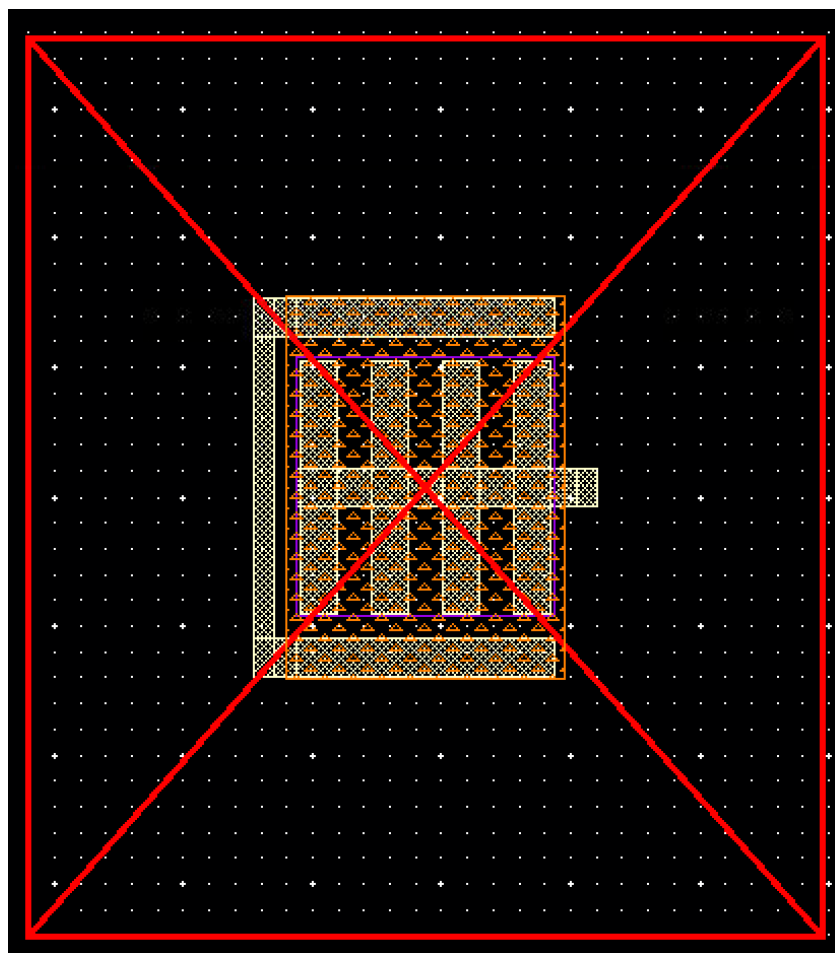
Attribute	Connectivity	Parameter	Property	ROD
Model name		crtmom		
description		CRTMOM		
fingers Width(M)		100n		
fingers Space(M)		100n		
Number of horizontal fingers		32		
Number of vertical fingers		32		
RTMOM Bottom Metal Layer		1		
RTMOM Top Metal Layer		7		
CapValue@0V_(F)		140.154f		
Create_OD_dummy		☒		
OD_width(M)		100n		
Create_OD2_Layer		☐		
DFM_rule		☐		
WellType		N		
Adding_DMEXCL_Layer		☒		
Hard_constrain		☒		
Register_number		Tsinghua-2214524229		

上图为MOM电容版图的参数变化；

1.9、版图器件介绍MIM电容 (tsmcN65)

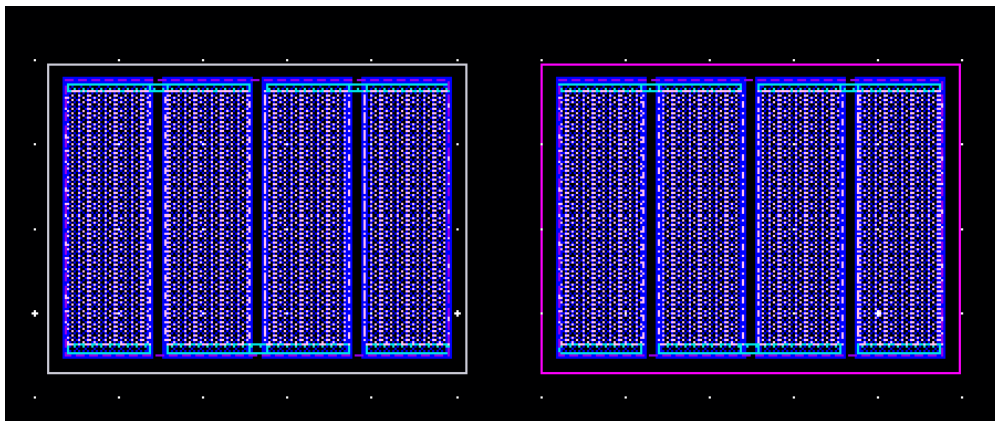


左图展示了MIM电容的版图绘制方法；右图为MIM电容的版图参数设置；

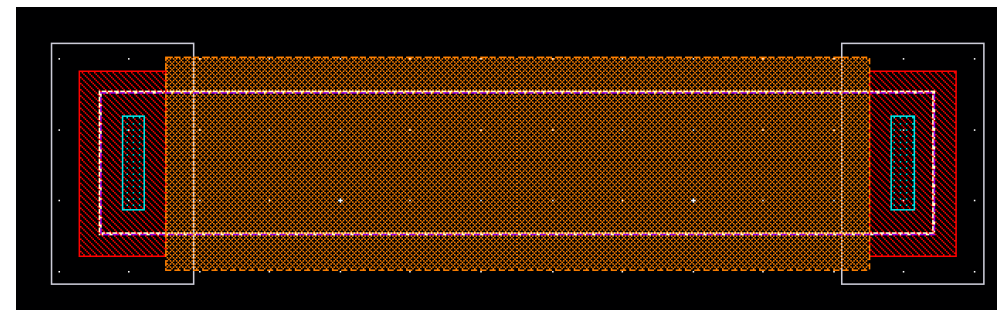


Attribute	Connectivity	Parameter	Property	ROD
Model name		mimcap_sin		
select CAP		MIM_2.0fF		
Connect_Metal		8		
description		Base band MIM capacitor		
Entry_mode		I & w		
Capacitance@0V(F)		208.6f		
Total_Capacitance(F)		208.6f		
Length_(M)		10u		
Width_(M)		10u		
ArrayX		1		
ArrayY		1		
External_metal_connect		☒		
Hard_constrain		☒		
☐ Display CDF Parameter Name				

1.10、版图器件介绍电阻 (tsmcN65)



上图为两种类型的poly电阻，左为做在NN里面的poly电阻，右为做在PP里的poly电阻；

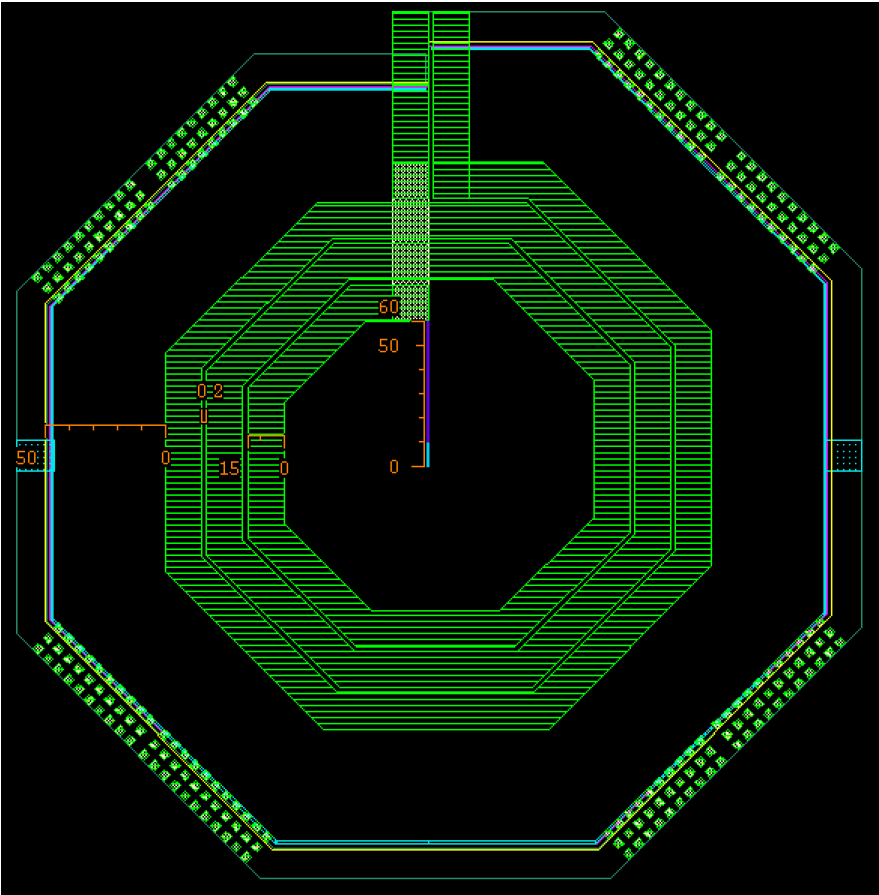


上图为N阱电阻，是一种利用n型阱结构作为电阻主体的电阻器；

1.11、版图器件介绍电感-1 (tsmcN65)



下图展示了电感：“spiral_std_mu_z” 的版图和参数设置；

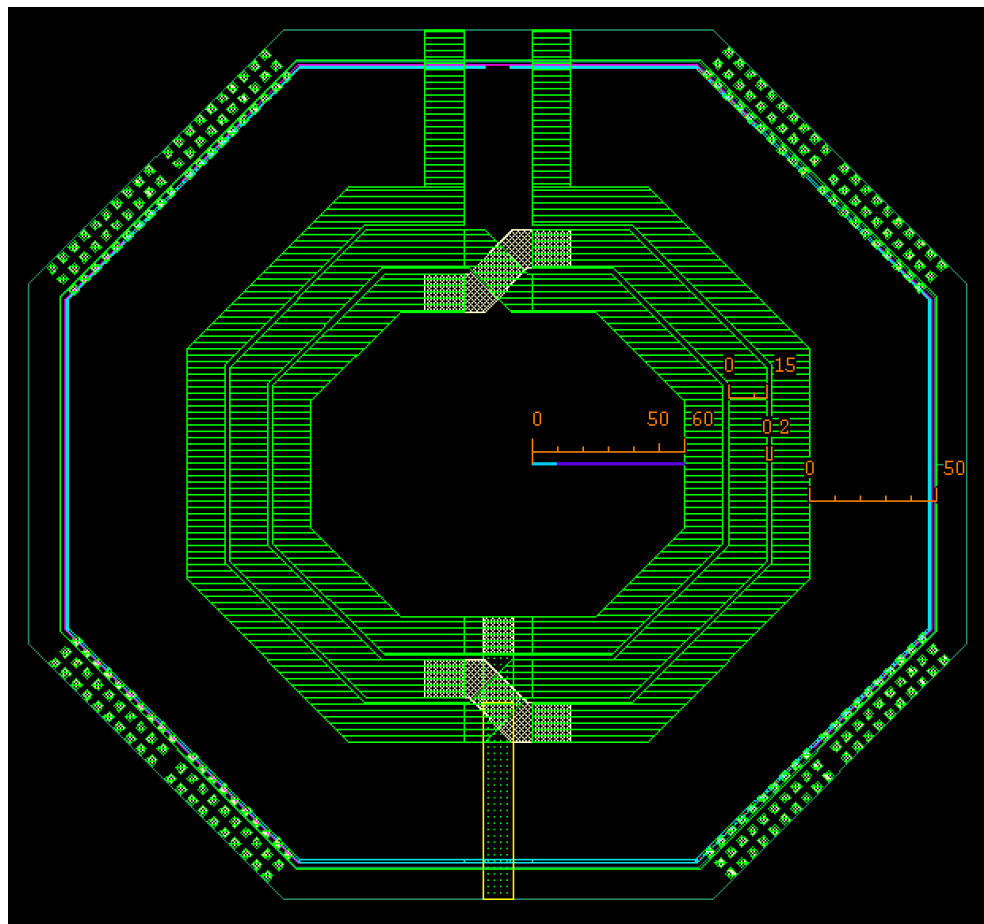


Attribute	Connectivity	Parameter	Property	ROD
Model name		spiral_std_mu_z		
Inductor_Width(M)		15.0u		
Inner_Radius(M)		60u		
Number_Of_Turns		3		
Spacing_(M)		2u		
Guard_Ring_Distances_(M)		50u		
temp(C)		27		
freq(Hz)		2.4G		
Approx. inductance(H)		2.17729n		
Q_factor		1.471445e+01		
Inductor_area_width(M)		359.00u		
Inductor_area_length(M)		350.5u		
TopMetal		9		
Hard_constrain		☒		
☐ Display CDF Parameter Name				

1.12、版图器件介绍电感-2 (tsmcN65)



下图展示了电感：“spiral_sym_ct_mu_z” 的版图和参数设置；

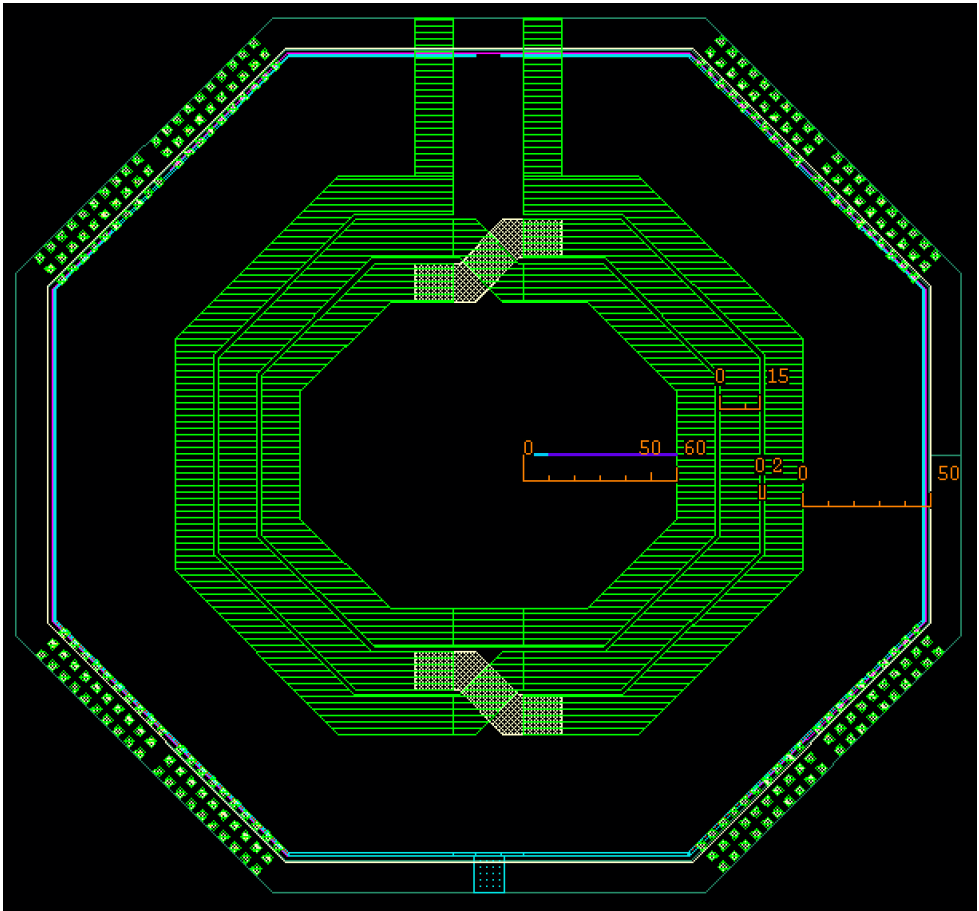


Attribute	Connectivity	Parameter	Property	ROD
Model name	spiral_sym_ct_mu_z			
Inductor_Width(M)	15u			
Inner_Radius(M)	60u			
Number_Of_Turns	3			
Spacing_(M)	2u			
Guard_Ring_Distances_(M)	50u			
temp(C)	27			
freq(Hz)	2.4G			
Approx. inductance(H)	2.24105n			
Q_factor	16.73677			
Inductor_area_width(M)	369.23u			
Inductor_area_length(M)	342.02u			
TopMetal	9			
Hard_constrain	☒			
☐ Display CDF Parameter Name				

1.13、版图器件介绍电感-3 (tsmcN65)

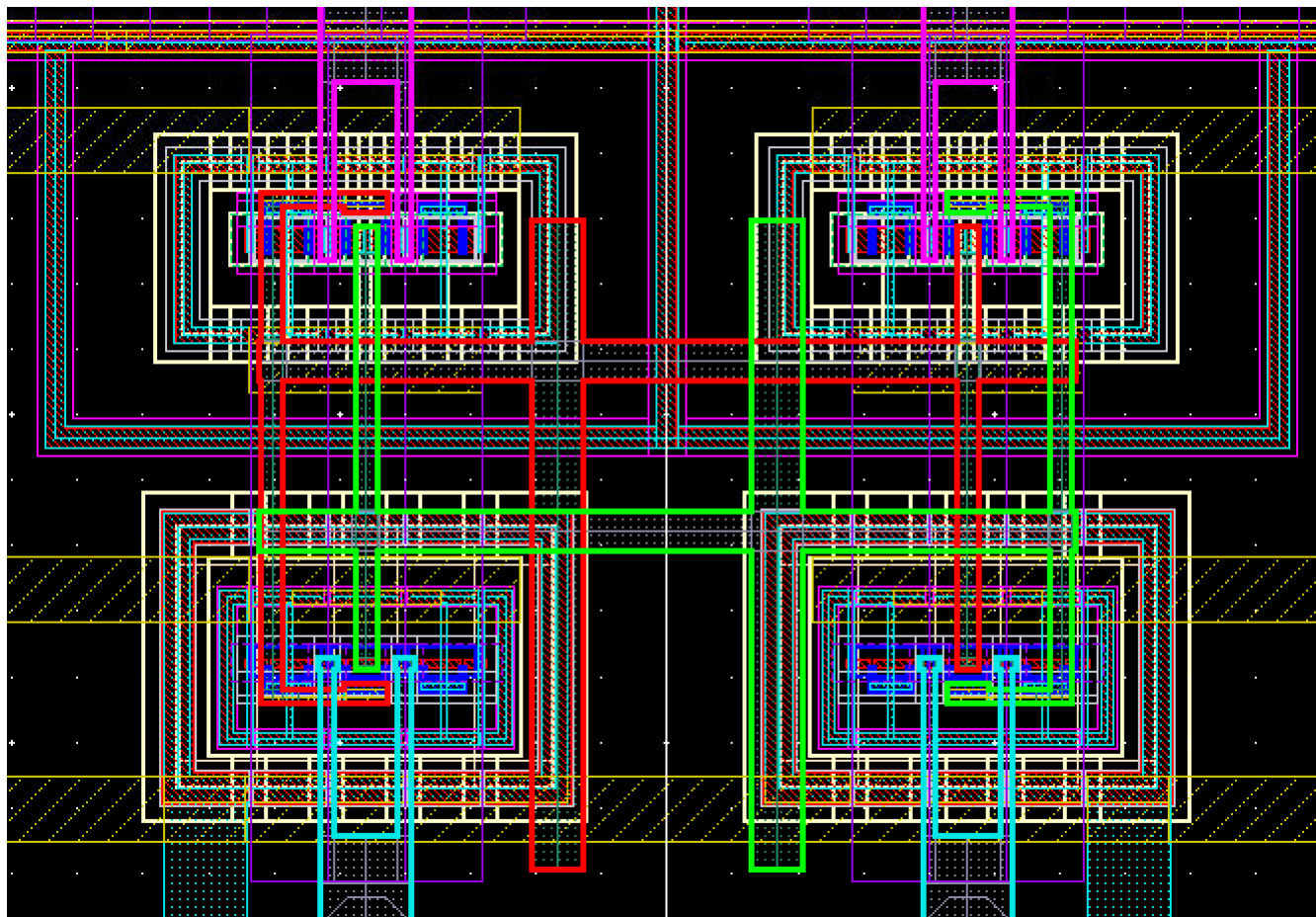


下图展示了电感：“spiral_sym_mu_z” 的版图和参数设置；



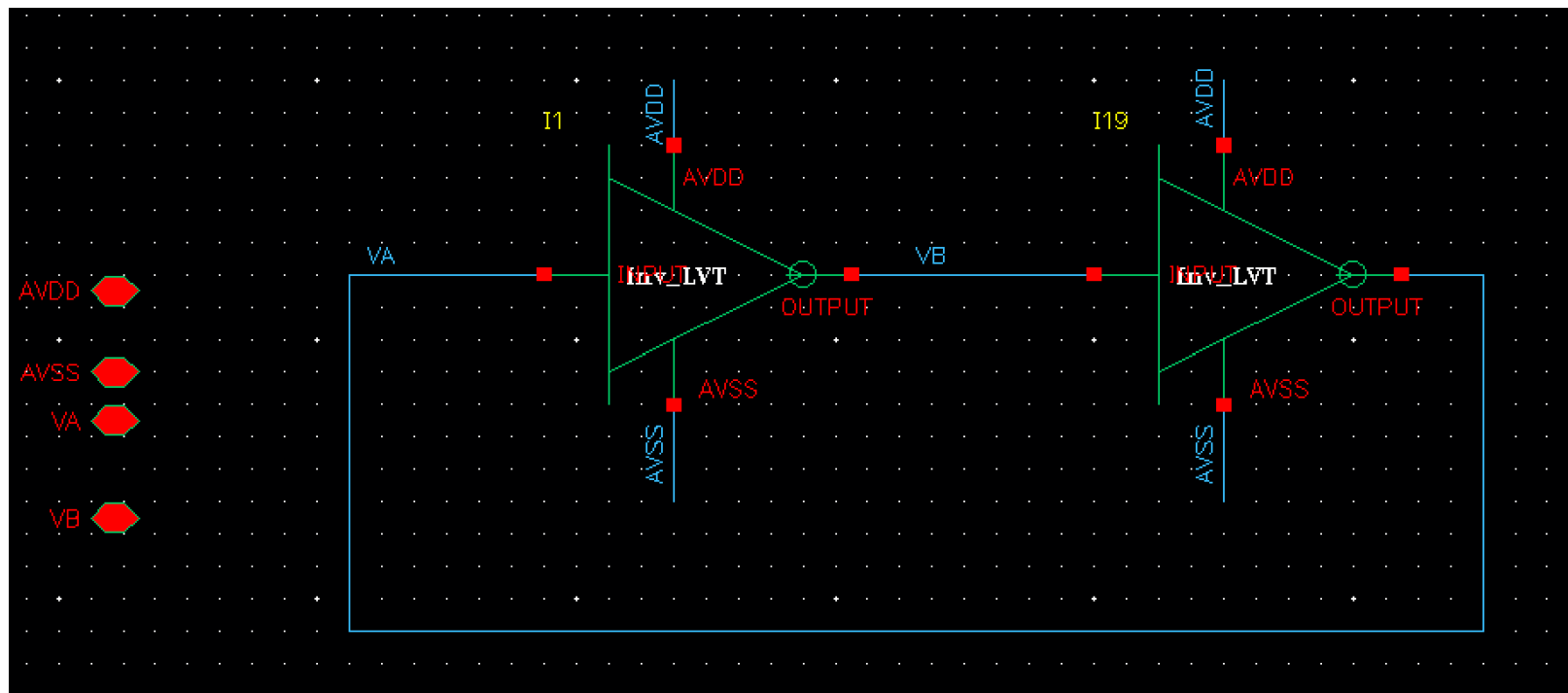
Attribute	Connectivity	Parameter	Property	ROD
Model name		spiral_sym_mu_z		
Inductor_Width(M)		15u		
Inner_Radius(M)		60u		
Number_Of_Turns		3		
Spacing_(M)		2u		
Guard_Ring_Distances_(M)		50u		
temp(C)		27		
freq(Hz)		2.4G		
L_value_single_ended(H)		2.28707n		
Q_factor_single_ended		14.67827		
Inductor_area_width(M)		369.23u		
Inductor_area_length(M)		342.02u		
TopMetal		9		
Hard_constrain		☒		
☐ Display CDF Parameter Name				

1.14、版图举例-1 (tsmcN65)



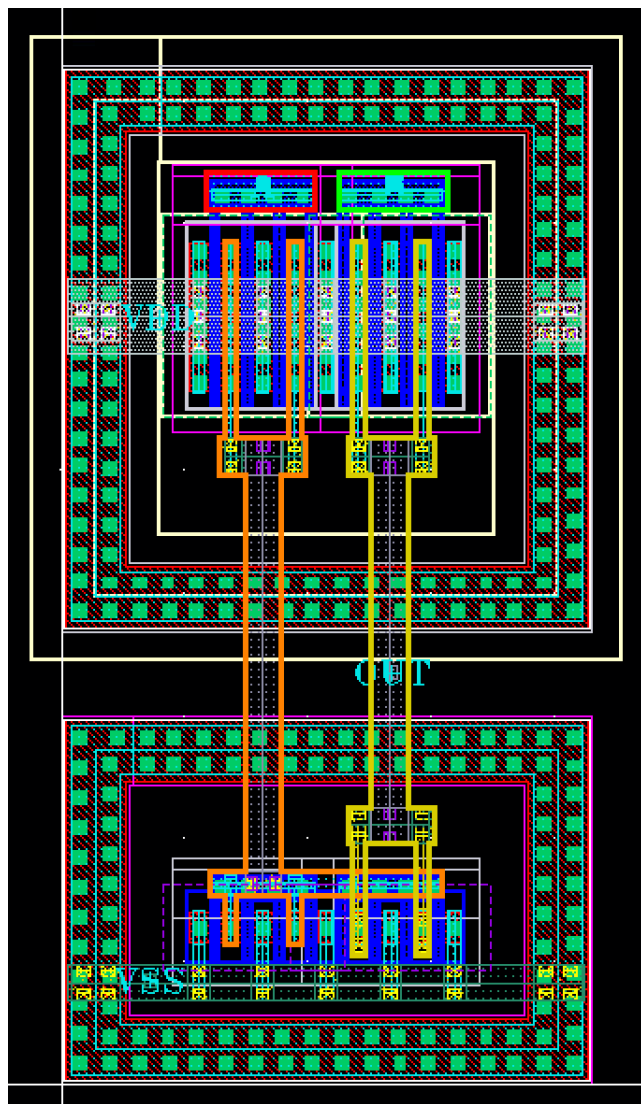
1、左图的版图大家可以思考一下电路是什么结构？
其中粉色线接VDD、蓝色线接VSS；

1.15、版图举例-1 (tsmcN65)



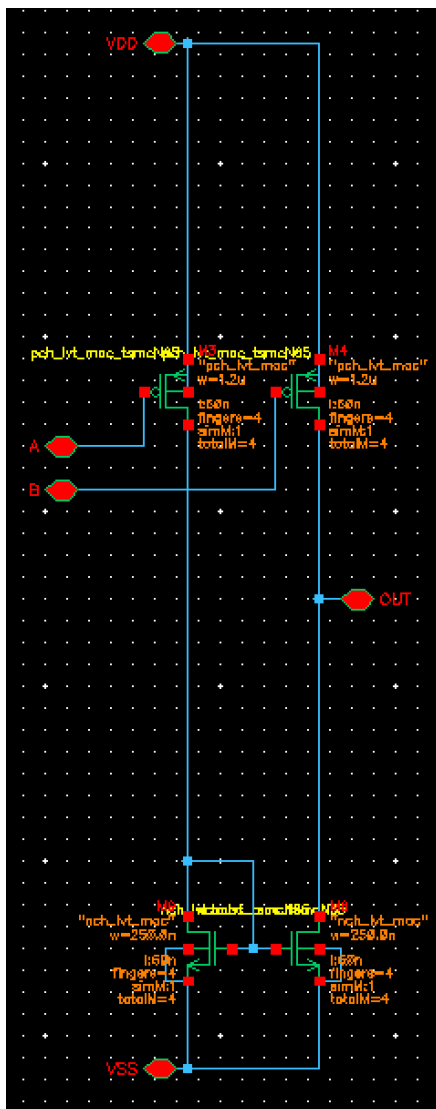
1、ES core;
Latch结构;

1.16、版图举例-2 (tsmcN65)



1、左图的版图大家可以思考一下电路是什么结构？

1.17、版图举例-2 (tsmcN65)

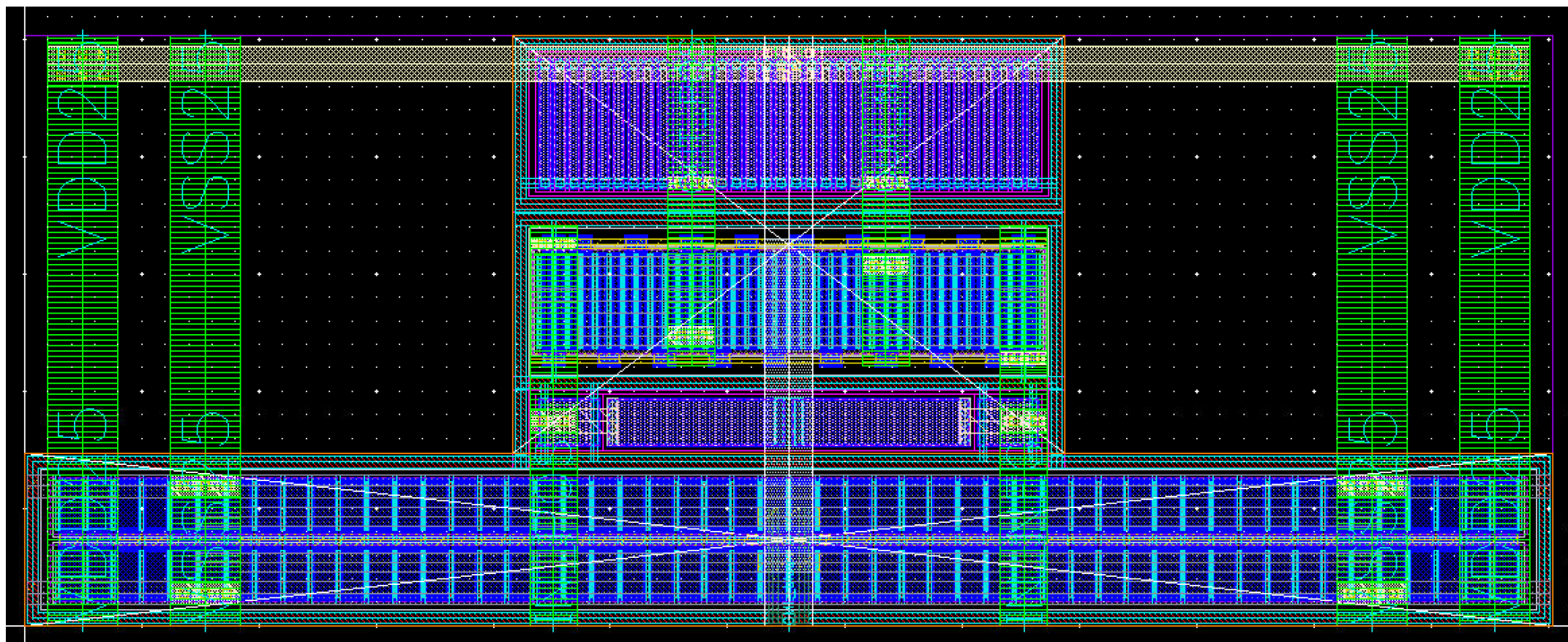


1、D2S结构;
差分输入转单端输出;

1.18、版图举例-3 (tsmcN65)

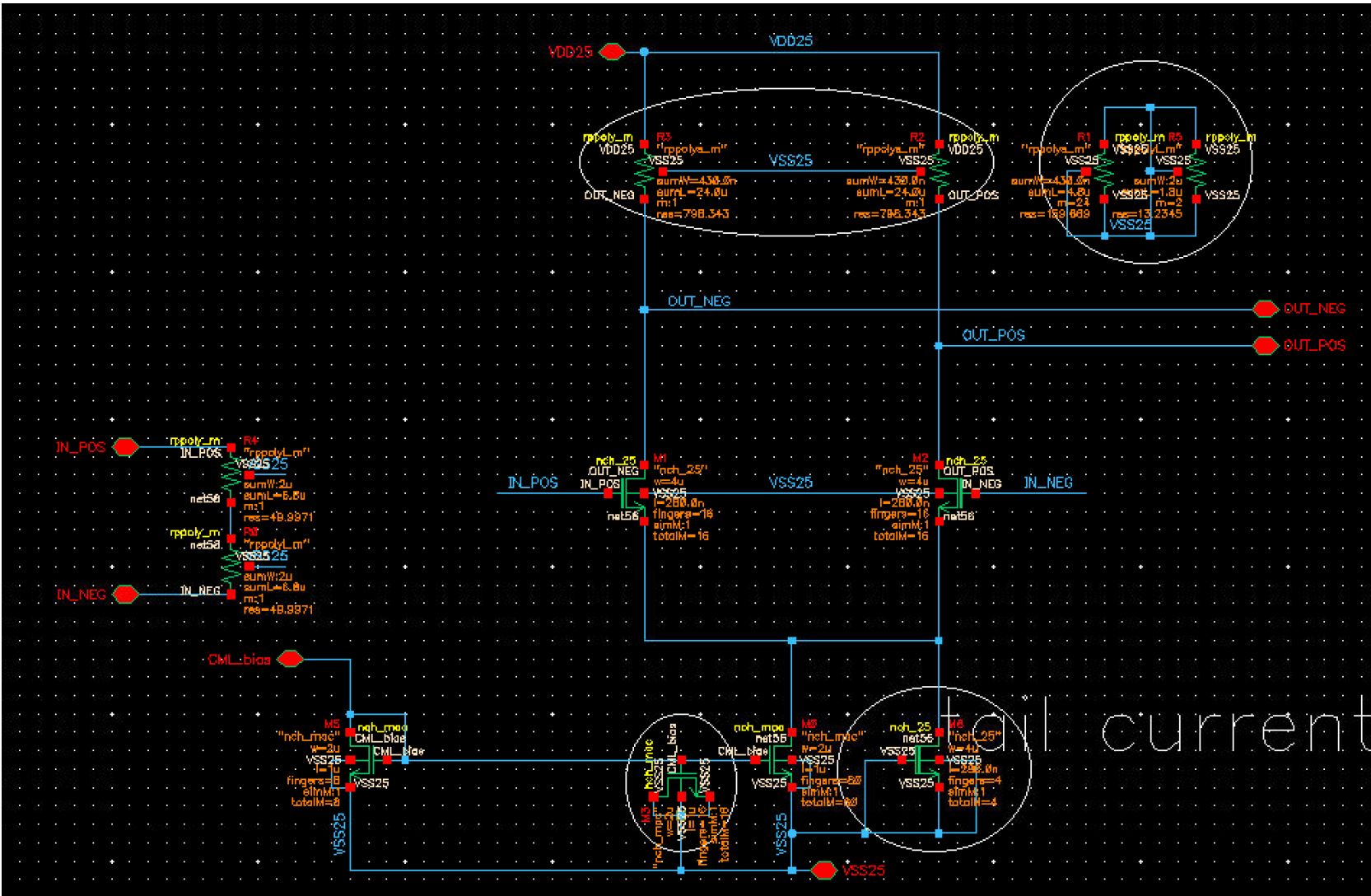


- 1、下图的版图为LVDS，目的是传输高速的差分信号；
- 2、版图较复杂，若不局部放大版图结构，较难区分；



1.19、版图举例-3 (tsmcN65)

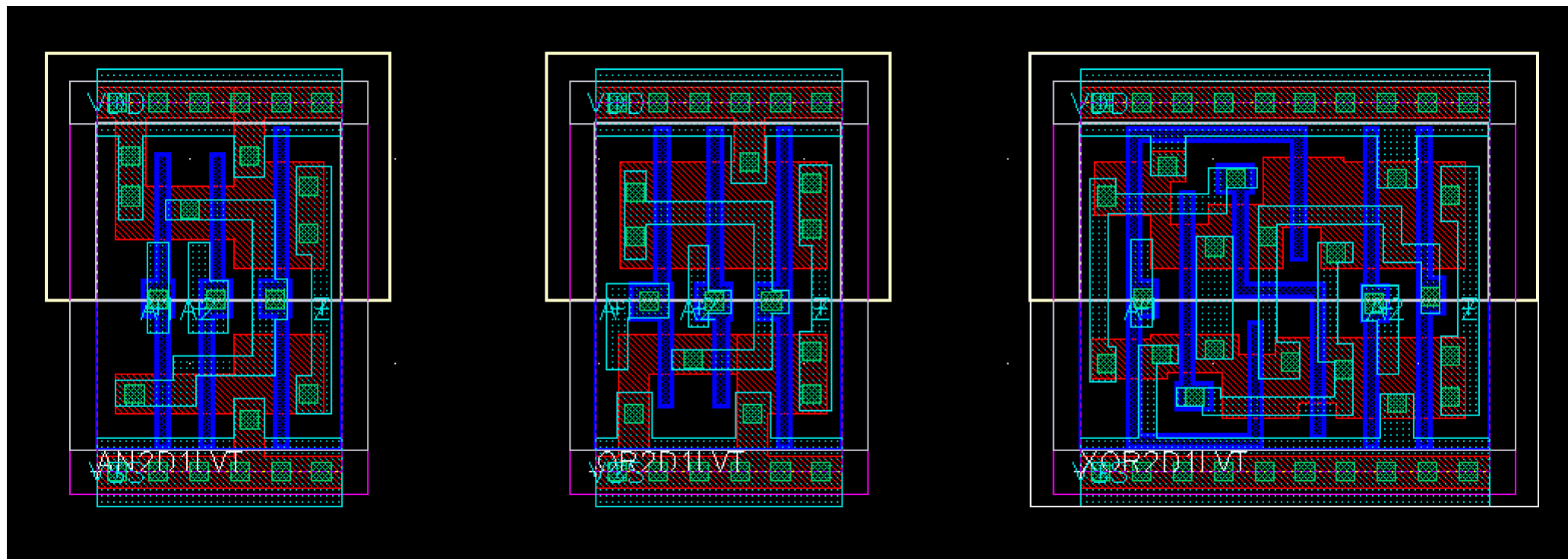
1、LVDS电路
如左图；



1.20、版图举例-4 (tsmcN65)



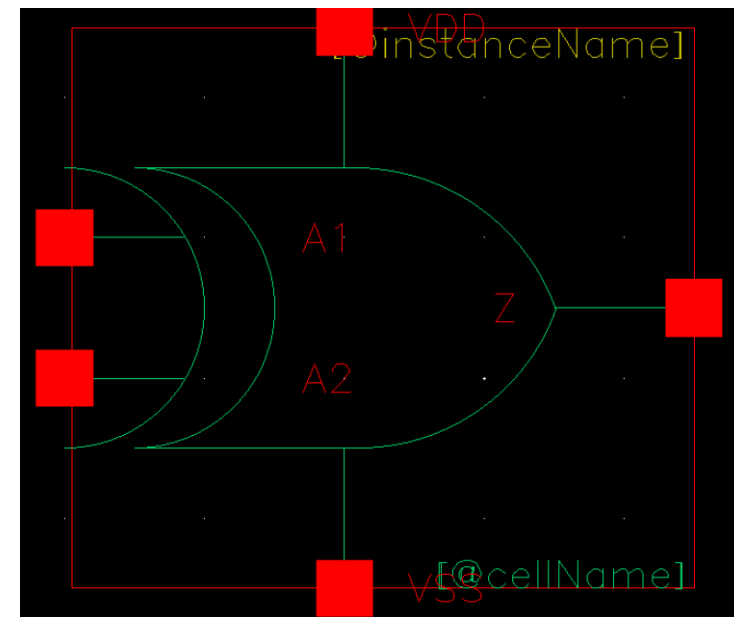
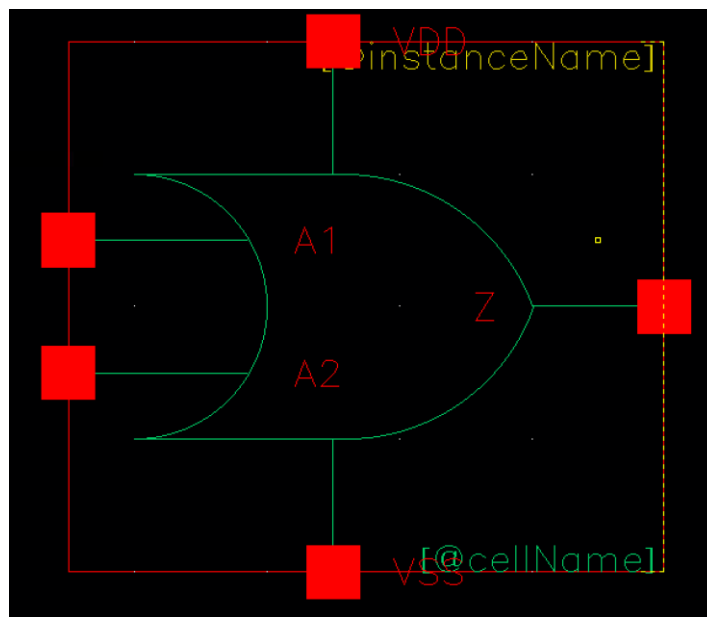
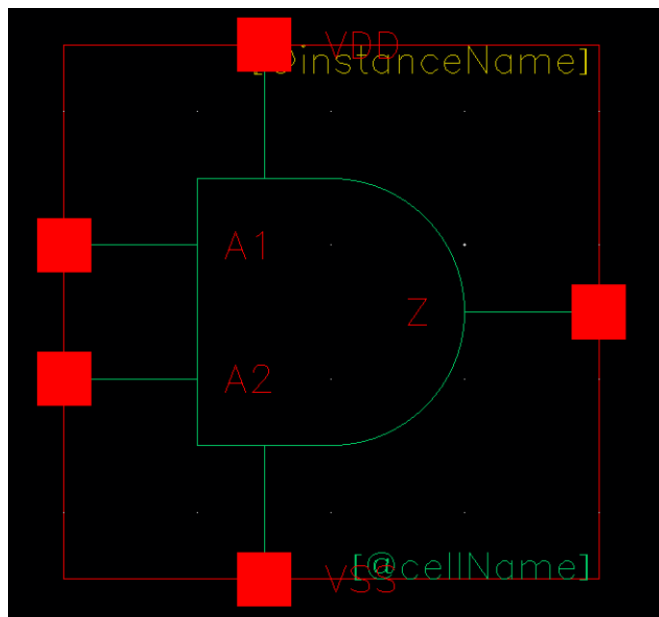
1、下图的版图大家可以思考一下电路是什么结构？



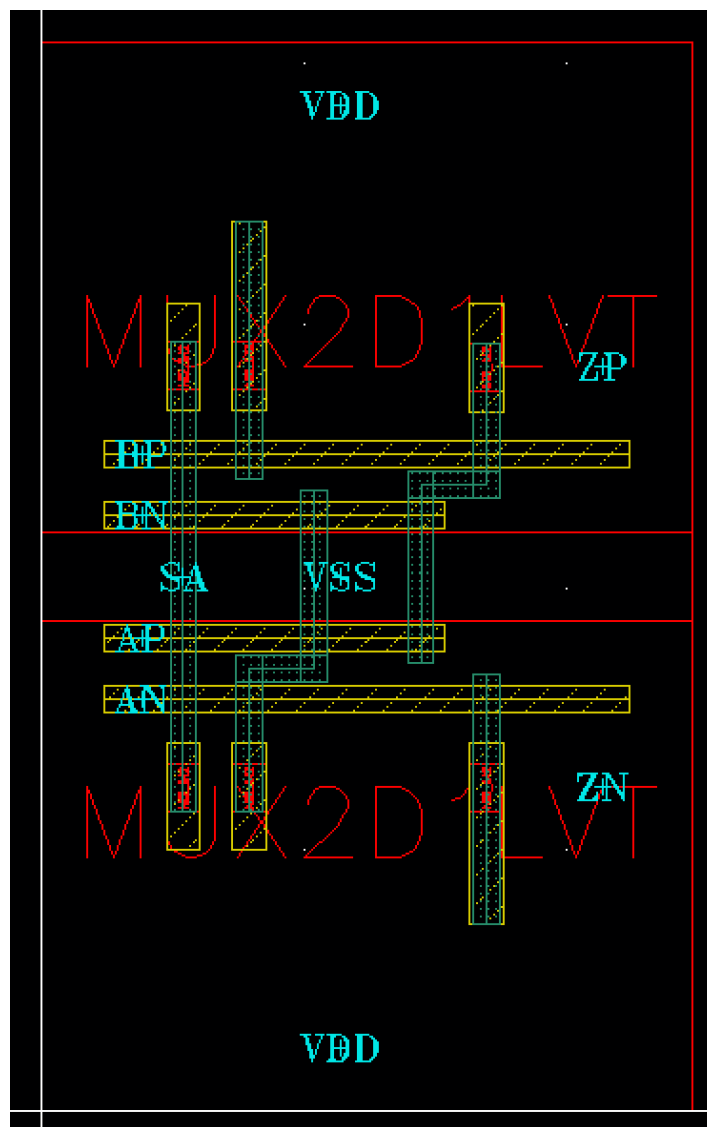
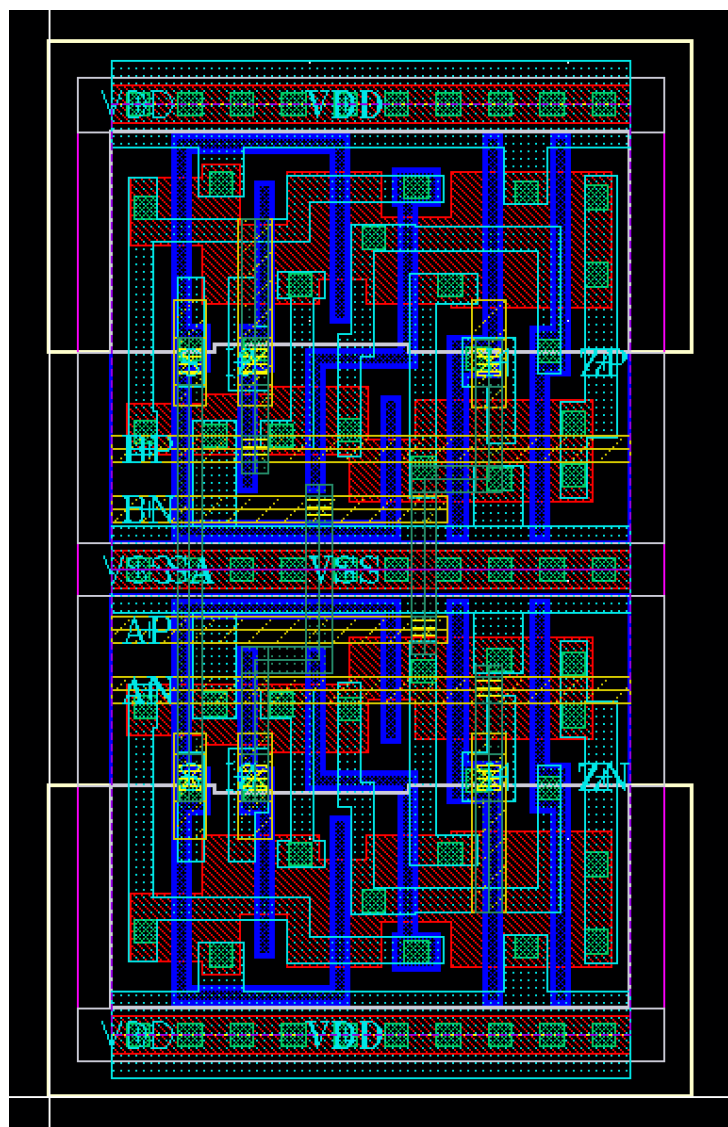
1.21、版图举例-4 (tsmcN65)



1、电路结构如下;



1.22、版图举例-5 (tsmcN65)

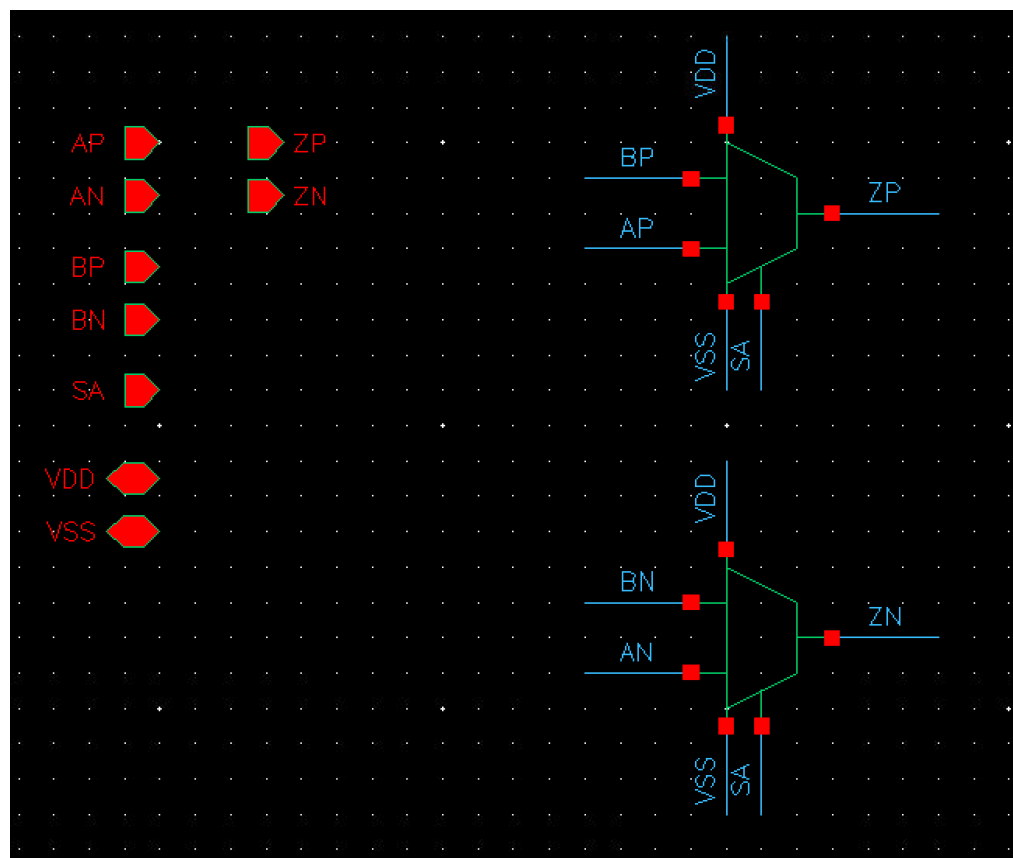


1、四路输入两路输出的多路选择器版图;

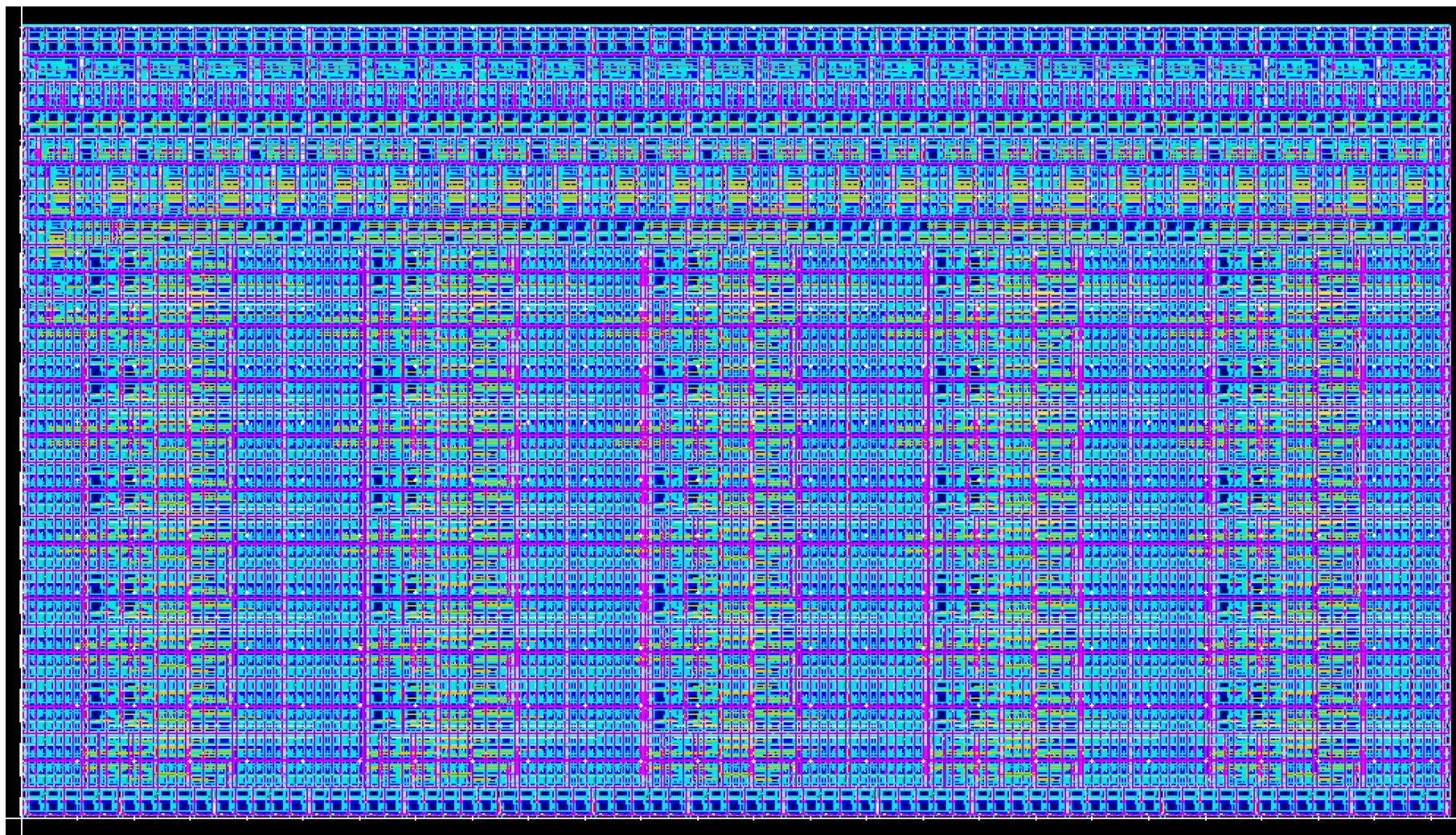
1.23、版图举例-5 (tsmcN65)



1、四路输入两路输出的多路选择器电路；



1.24、版图举例-6 (tsmcN65)



1、用标准单元进行一定规模的版图绘制；



谢谢大家！

28nm 工艺介绍

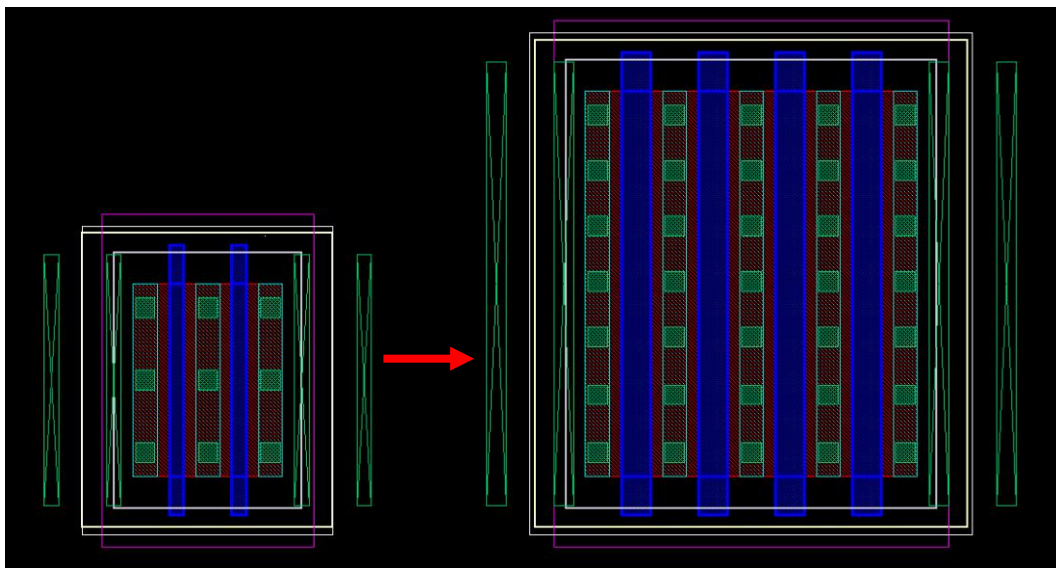


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School of Integrated Circuits, Tsinghua University

■附件为28nm相关器件版图介绍。



1.1、版图器件介绍PMOS (tsmcN28)

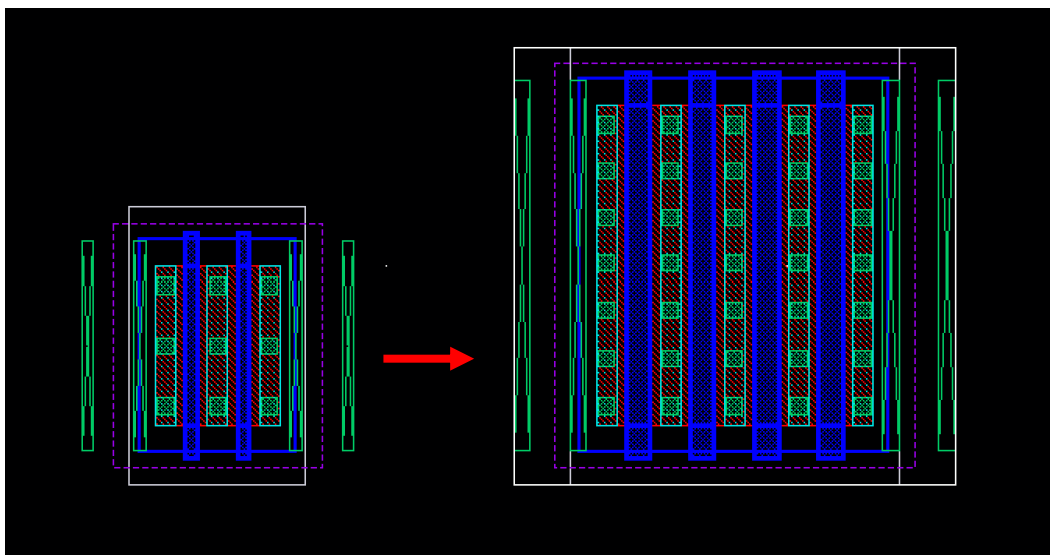


上图PMOS版图，当版图尺寸
由 $L=30n$ 、 $W=400n$ 、 $Fingers=2$ ；
改为 $L=60n$ 、 $W=800n$ 、 $Fingers=4$ ；
版图的变化亦如上图所示；
右图为版图的参数设置；

Attribute	Connectivity	Parameter	Property	ROD
Model name		pch_lvt_mac		
description		core volt LOW VT PMOS transistor		
Length(M)		30n		
Finger_width(M)		400n		
Width(M)		800n		
Number of Fingers		2		
total_m		2		
Hard_constrain		☒		
Previous pre-set		Preset1		
pre-set-Option		Custom		
Display S/D Parameters		☒ OFF	☐ ON	
Display LOD Effect		☒ OFF	☐ ON	
Display Well Proximity Effect		☒ OFF	☐ ON	
Display Poly Spacing Effect		☒ OFF	☐ ON	
Display OD-OD Spacing Effect		☒ OFF	☐ ON	
Display MBE Effect		☒ OFF	☐ ON	
Display Device Characteristics		☒ OFF	☐ ON	
Display DFM Options		☒ OFF	☐ ON	
Display Layout Related Parameters		☒ OFF	☐ ON	

Attribute	Connectivity	Parameter	Property	ROD
Model name		pch_lvt_mac		
description		core volt LOW VT PMOS transistor		
Length(M)		60n		
Finger_width(M)		800n		
Width(M)		3.2u		
Number of Fingers		4		
total_m		4		
Hard_constrain		☒		
Previous pre-set		Preset1		
pre-set-Option		Custom		
Display S/D Parameters		☒ OFF	☐ ON	
Display LOD Effect		☒ OFF	☐ ON	
Display Well Proximity Effect		☒ OFF	☐ ON	
Display Poly Spacing Effect		☒ OFF	☐ ON	
Display OD-OD Spacing Effect		☒ OFF	☐ ON	
Display MBE Effect		☒ OFF	☐ ON	
Display Device Characteristics		☒ OFF	☐ ON	
Display DFM Options		☒ OFF	☐ ON	
Display Layout Related Parameters		☒ OFF	☐ ON	

1.2、版图器件介绍NMOS (tsmcN28)



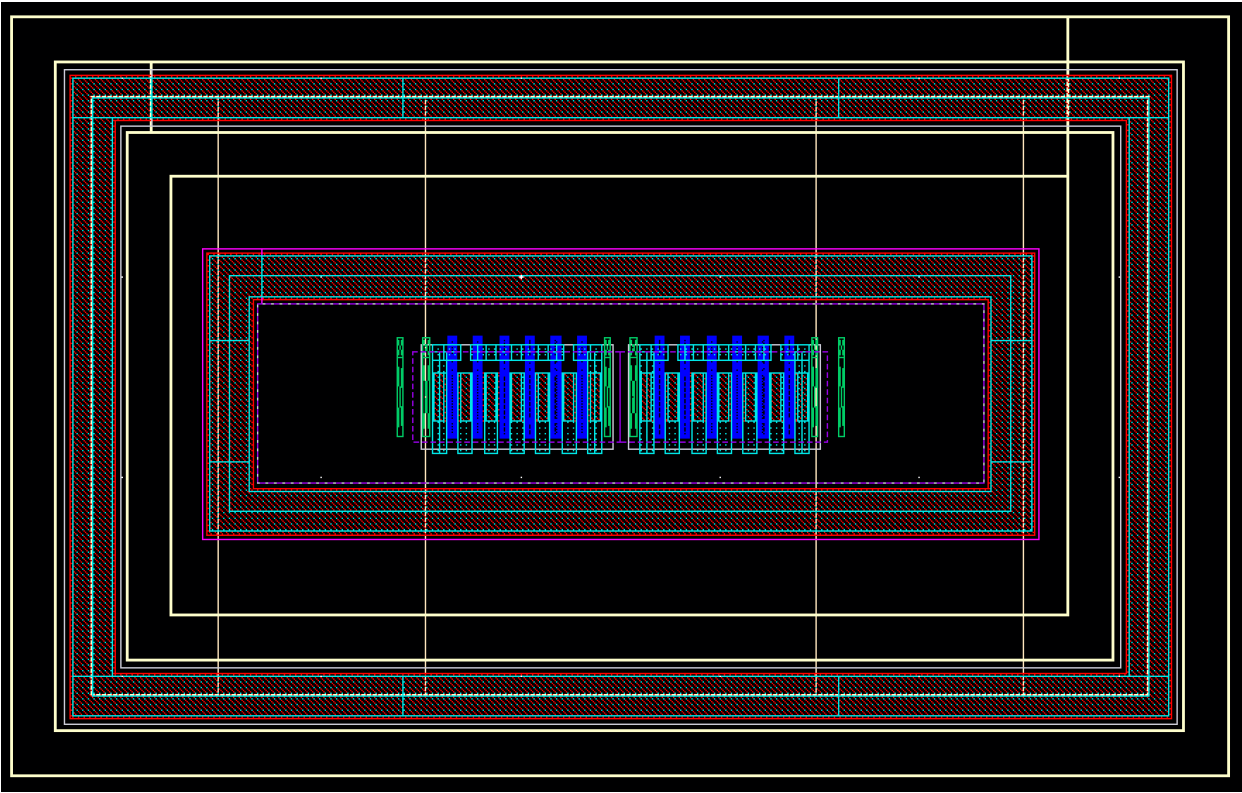
上图为NMOS版图，当版图尺寸由L=30n、W=400n、Fingers=2；改为L=60n、W=800n、Fingers=4；版图的变化亦如上图所示；

右图为版图的参数设置；

Attribute	Connectivity	Parameter	Property	ROD
Model name		nch_lvt_mac		
description		core volt LOW VT NMOS transistor		
Length(M)		30n		
Finger_width(M)		400n		
Width(M)		800n		
Number of Fingers		2		
total_m		2		
Hard_constrain		☒		
Previous pre-set		Preset1		
pre-set-Option		Custom		
Display S/D Parameters		☒ OFF	☐ ON	
Display LOD Effect		☒ OFF	☐ ON	
Display Well Proximity Effect		☒ OFF	☐ ON	
Display Poly Spacing Effect		☒ OFF	☐ ON	
Display OD-OD Spacing Effect		☒ OFF	☐ ON	
Display MBE Effect		☒ OFF	☐ ON	
Display Device Characteristics		☒ OFF	☐ ON	
Display DFM Options		☒ OFF	☐ ON	
Display Layout Related Parameters		☒ OFF	☐ ON	

Attribute	Connectivity	Parameter	Property	ROD
Model name		nch_lvt_mac		
description		core volt LOW VT NMOS transistor		
Length(M)		60n		
Finger_width(M)		800n		
Width(M)		3.2u		
Number of Fingers		4		
total_m		4		
Hard_constrain		☒		
Previous pre-set		Preset1		
pre-set-Option		Custom		
Display S/D Parameters		☒ OFF	☐ ON	
Display LOD Effect		☒ OFF	☐ ON	
Display Well Proximity Effect		☒ OFF	☐ ON	
Display Poly Spacing Effect		☒ OFF	☐ ON	
Display OD-OD Spacing Effect		☒ OFF	☐ ON	
Display MBE Effect		☒ OFF	☐ ON	
Display Device Characteristics		☒ OFF	☐ ON	
Display DFM Options		☒ OFF	☐ ON	
Display Layout Related Parameters		☒ OFF	☐ ON	

1.3、版图器件介绍DNW_NMOS (tsmcN28)



Apply To:

Show: ☐ system ☒ user ☒ CDF

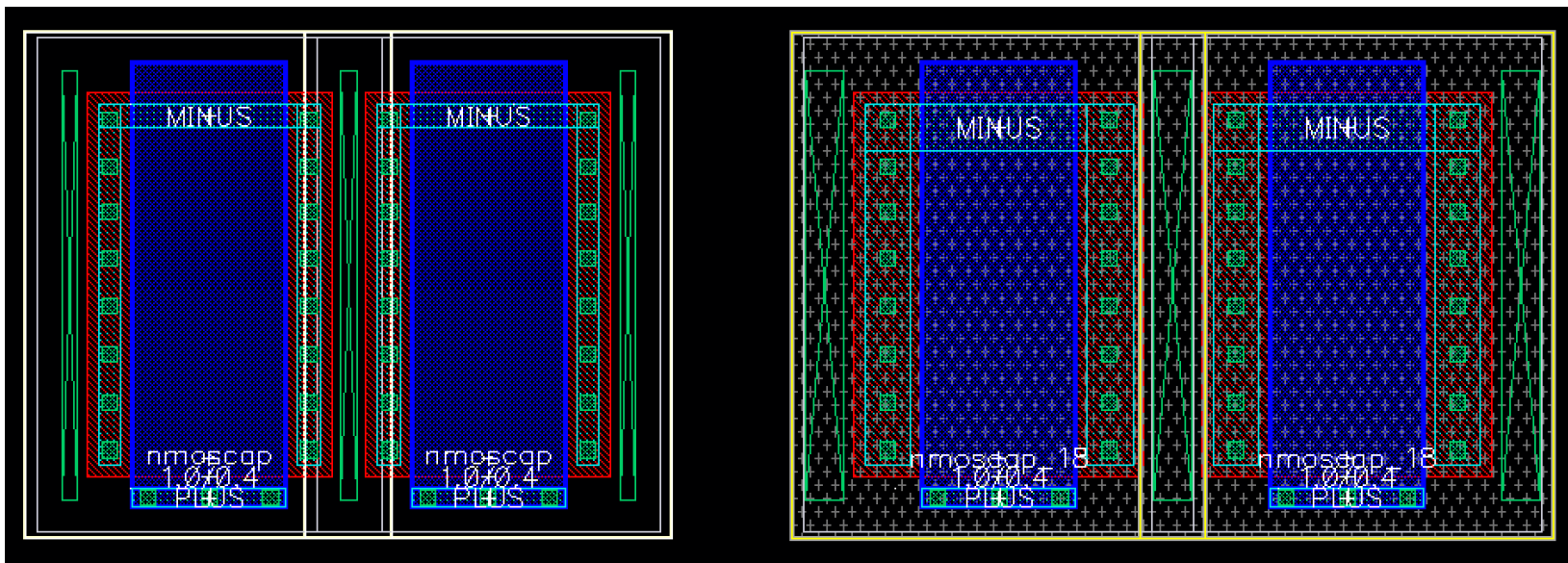
Property	Value	Display
Library Name	tsmcN28	off
Cell Name	nch_dnw_mac	value
View Name	symbol1	off
Instance Name	M0	off

CDF Parameter	Value	Display
Model name	nch_mac	off
description	volt standard VT NMOS in DNW transistor	off
Length(M)	30n M	off
Finger_width(M)	240.0n M	off
Width(M)	2.88u M	off
Number of Fingers	12	off
Multiplier	1	off
total_m	12	off
Hard_constrain	☒	off
Previous pre-set	Preset1	off
pre-set-Option	Custom	off
Display Simulation Flag	☒ OFF ☐ ON	off
Display Simulation Parameters	☒ OFF ☐ ON	off
Display S/D Parameters	☒ OFF ☐ ON	off
Display LOD Effect	☒ OFF ☐ ON	off

上图展示了DNW_NMOS的版图绘制方法;
右图为DNW_NMOS版图的参数设置;



1.4、版图器件介绍nmoscap/nmoscap18 (tsmcN28)

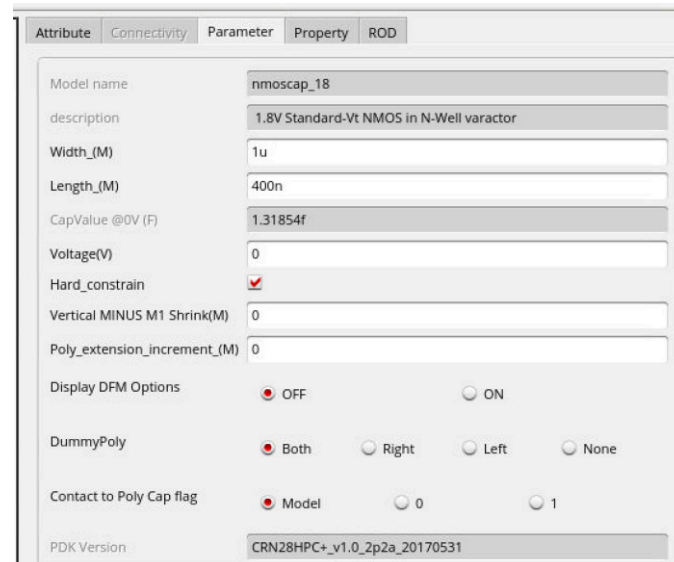
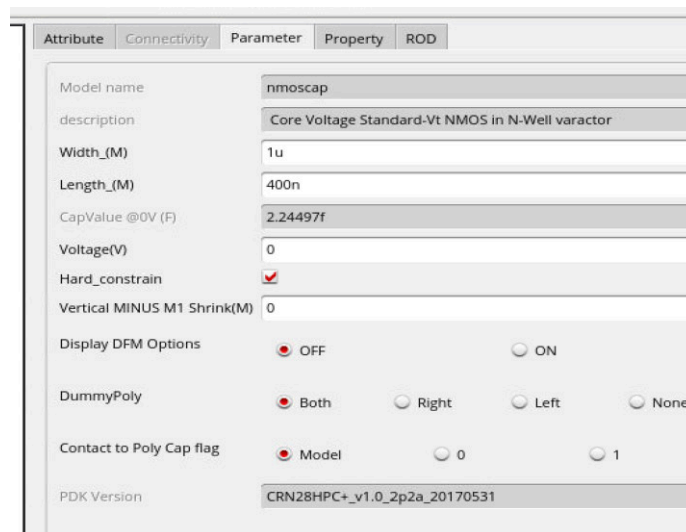


①nmoscap

②nmoscap18

上图展示了nmoscap和nmoscap18的版图绘制方法，二者的主要区别除了容值不同，nmoscap18的漏电比nmoscap要小，但二者都是广泛使用在集成电路中的mos电容；

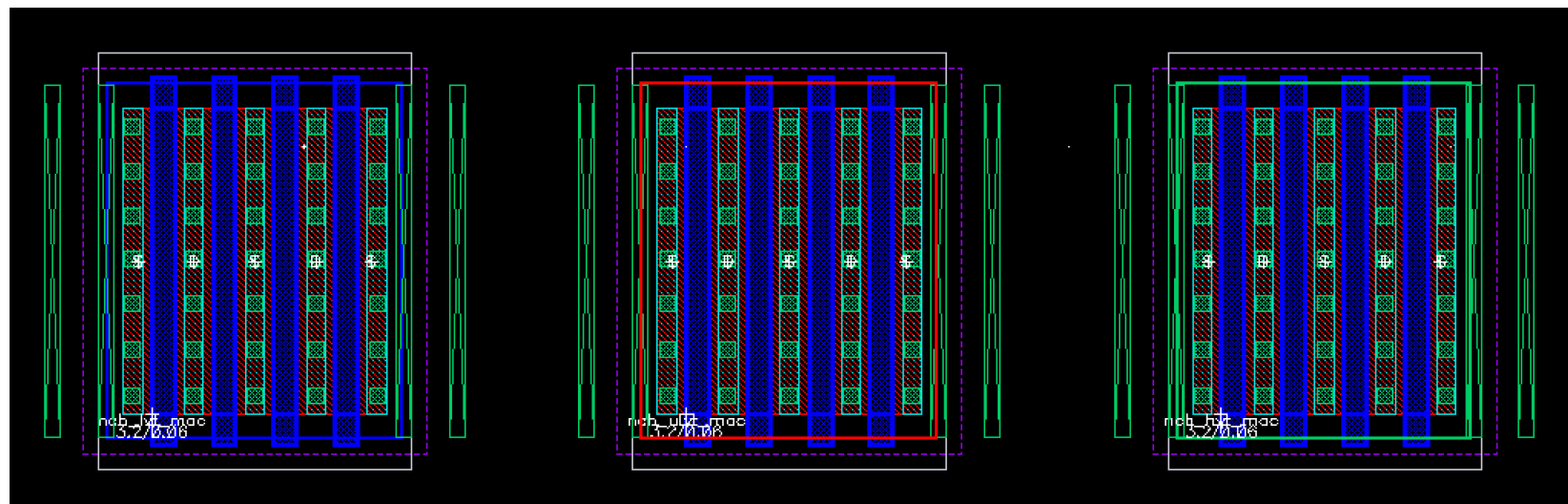
右图为nmoscap 的参数设置；



1.5、版图器件介绍lvt、 ulvt、 hvt (tsmcN28)



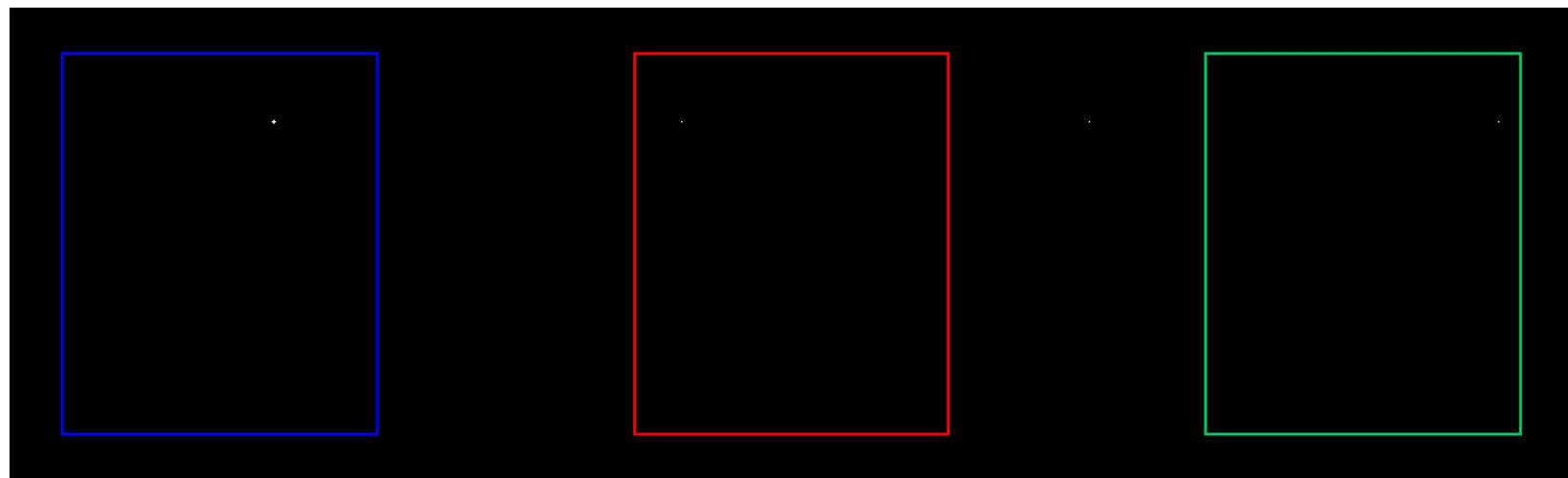
右图展示了lvt、ulvt、hvt三种类型的NMOS Core device，在版图上它们只有一层不同，而这一层即可以区分出它们的不同类型；



①lvt

②ulvt

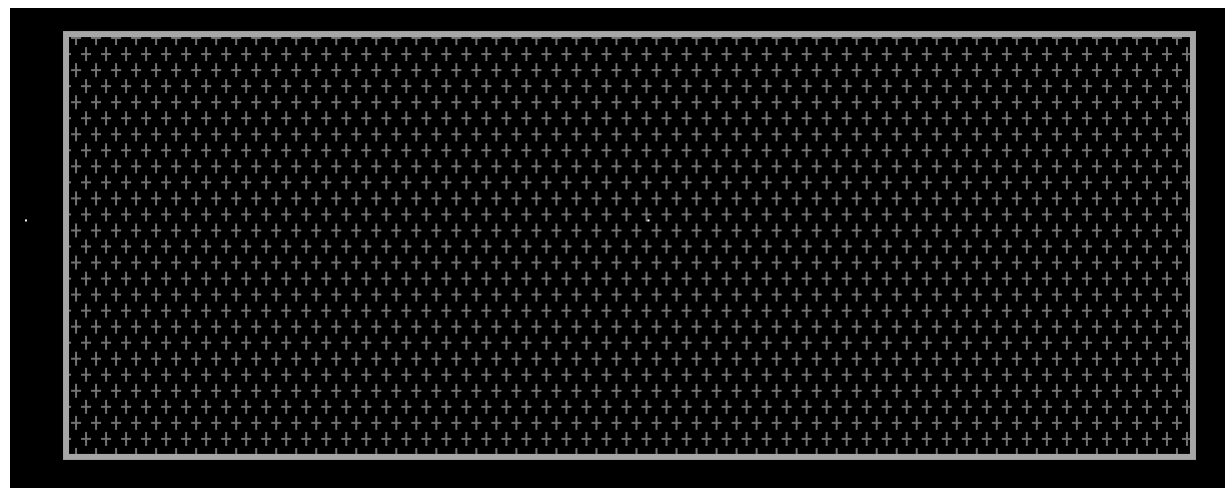
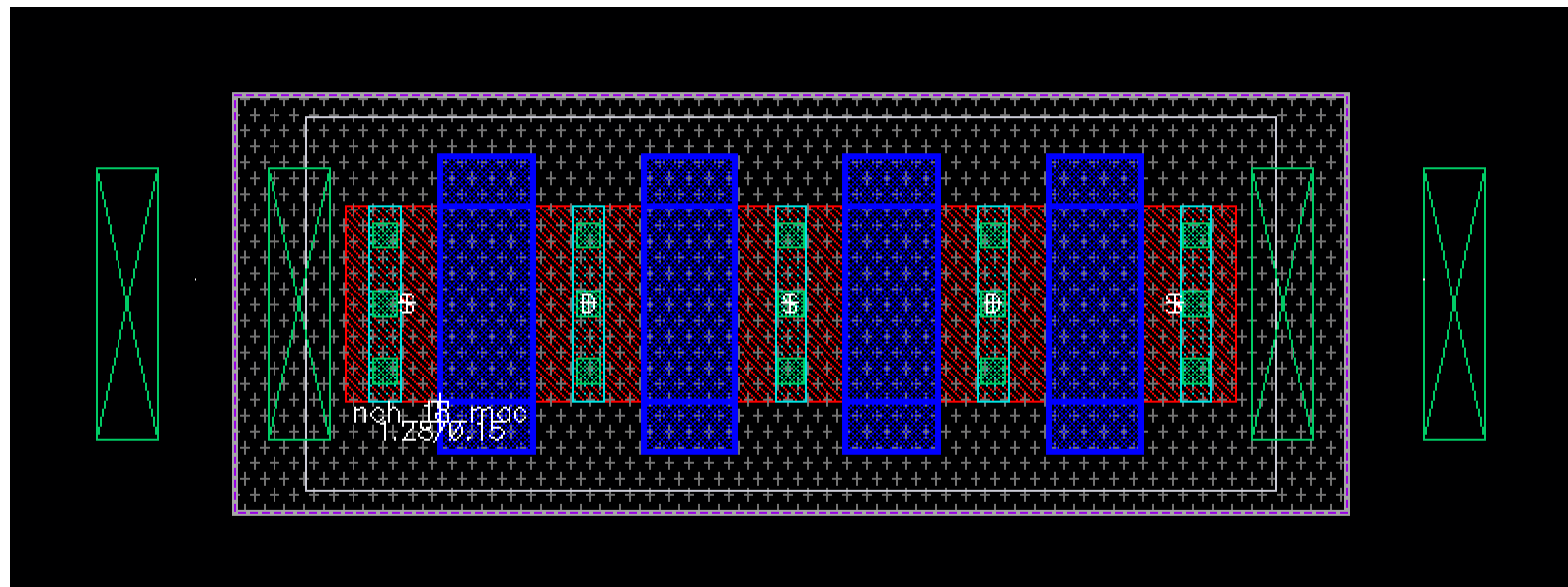
③hvt



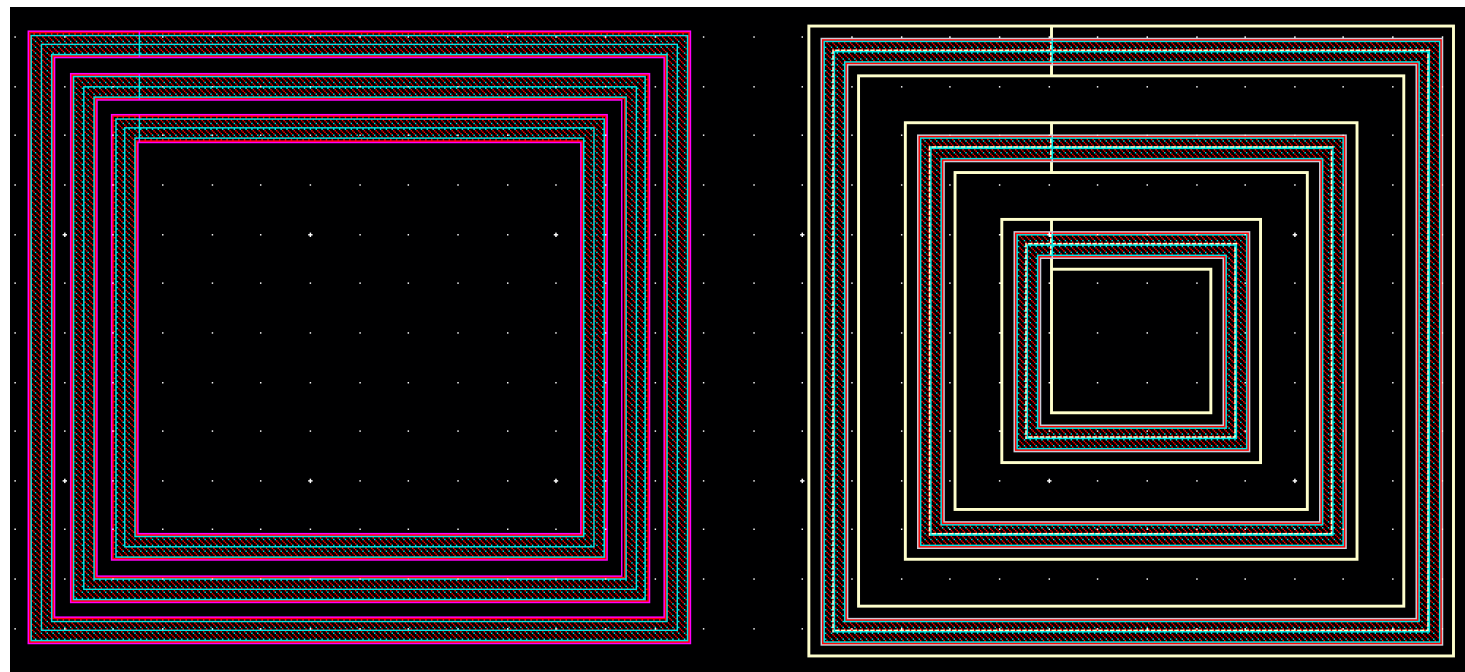
1.6、版图器件介绍1.8V (tsmcN28)



右图展示了1.8V类型的NMOS IO device, 同样, 与lvt、ulvt、hvt三种类型的NMOS Core device 相比, IO device用的是高压标识层进行区分;



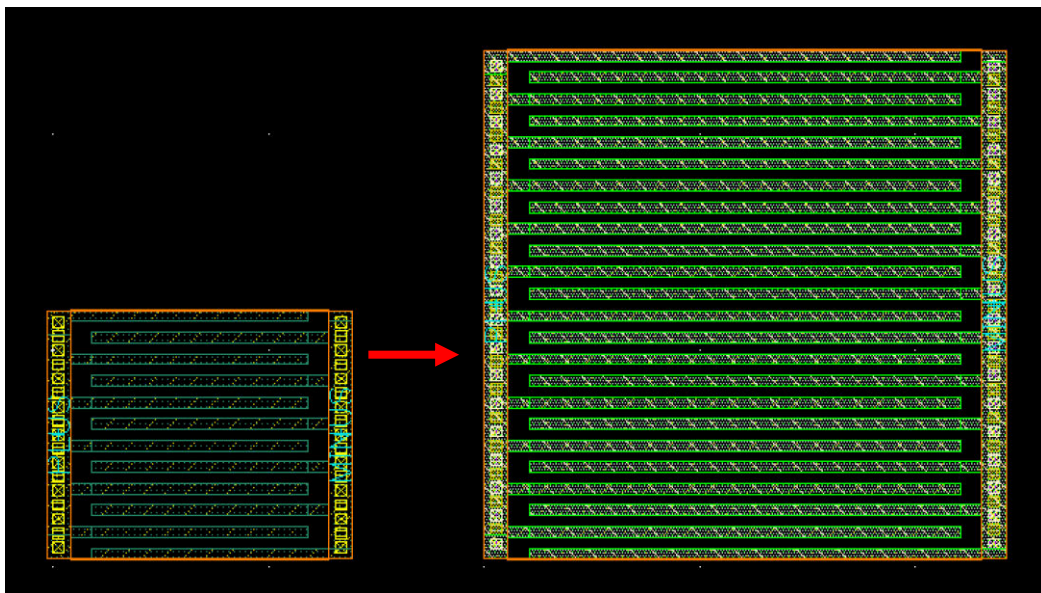
1.7、版图器件介绍GuardRing (tsmcN28)



上图为两种类型的GuardRing，分为Pwell（左）和Nwell（右），GuardRing的主要作用是作为P管和N管的衬底供电，以及对重要的版图模块进行保护等作用，也可以隔离噪声；



1.8、版图器件介绍MOM电容 (tsmcN28)



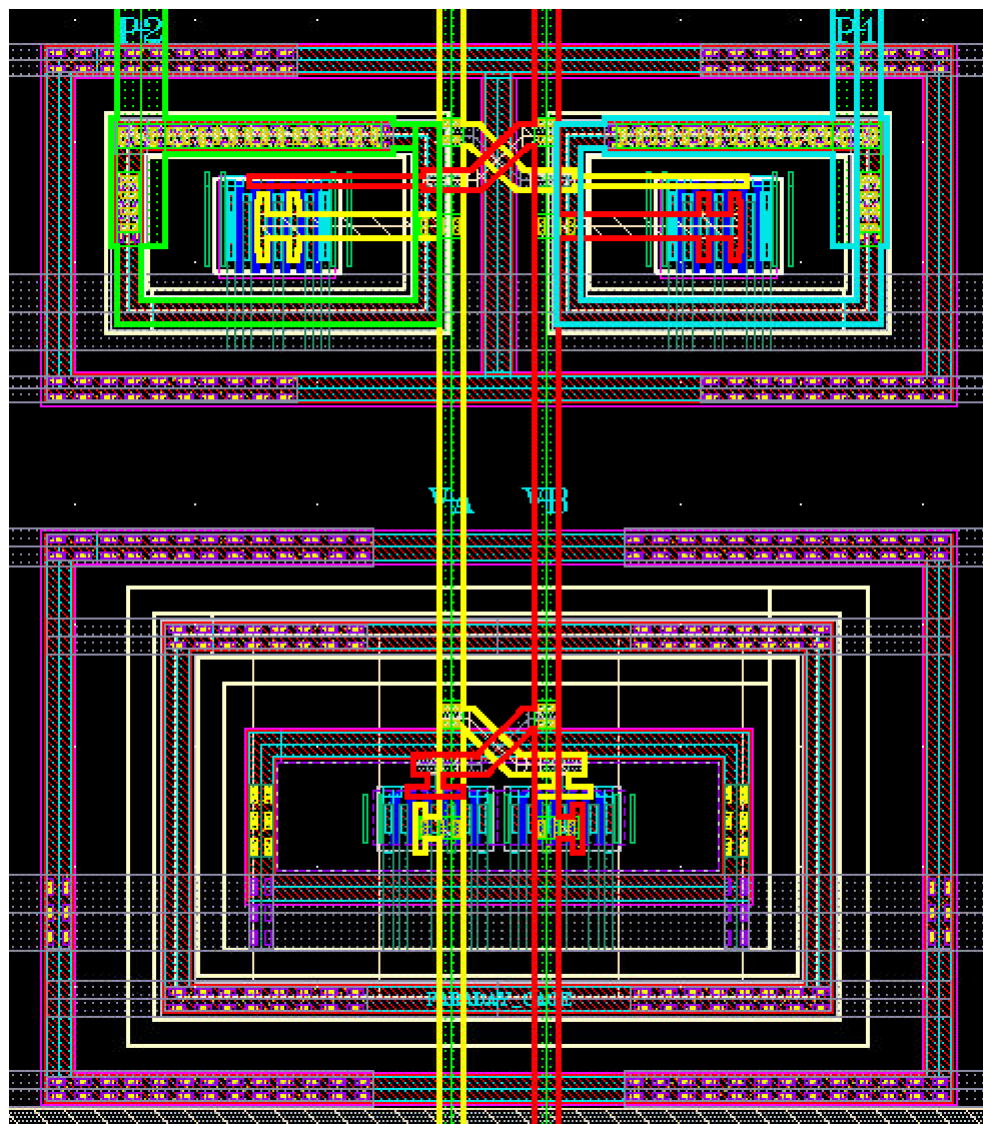
上图为MOM电容版图，当版图尺寸由 $W=50\text{n}$ 、 $S=50\text{n}$ 、 $L=1\mu$ 、 $\text{Finger}=12$ 、FMOM **BtoT**=M1-M3；改为 $W=50\text{n}$ 、 $S=50\text{n}$ 、 $L=2\mu$ 、 $\text{Finger}=24$ 、FMOM **BtoT** =M1-M7；版图的变化亦如上图所示；

Attribute	Connectivity	Parameter	Property	ROD
Model name		cfmom_2t		
description		CFMOM_2T		
fingers Width(M)		50n		
fingers Space(M)		50n		
fingers Length(M)		1u		
Number of fingers		12		
FMOM Bottom Metal Layer		1		
FMOM Top Metal Layer		3		
CapValue@0V_(F)		3.26053f		
Display_DMEXCL_and_DM_options		☐ ON ☒ OFF		
OD_POLY_Block		☐ ON ☒ OFF		
Hard_constrain		☒		
Register_number		SHICC-2028021446		
Sidewall area		2.97e-12		
PDK Version		CRN28HPC+_v1.0_2p2a_20170531		

Attribute	Connectivity	Parameter	Property	ROD
Model name		cfmom_2t		
description		CFMOM_2T		
fingers Width(M)		50n		
fingers Space(M)		50n		
fingers Length(M)		2u		
Number of fingers		24		
FMOM Bottom Metal Layer		1		
FMOM Top Metal Layer		7		
CapValue@0V_(F)		30.3453f		
Display_DMEXCL_and_DM_options		☐ ON ☒ OFF		
OD_POLY_Block		☐ ON ☒ OFF		
Hard_constrain		☒		
Register_number		SHICC-2028021446		
Sidewall area		2.898e-11		
PDK Version		CRN28HPC+_v1.0_2p2a_20170531		

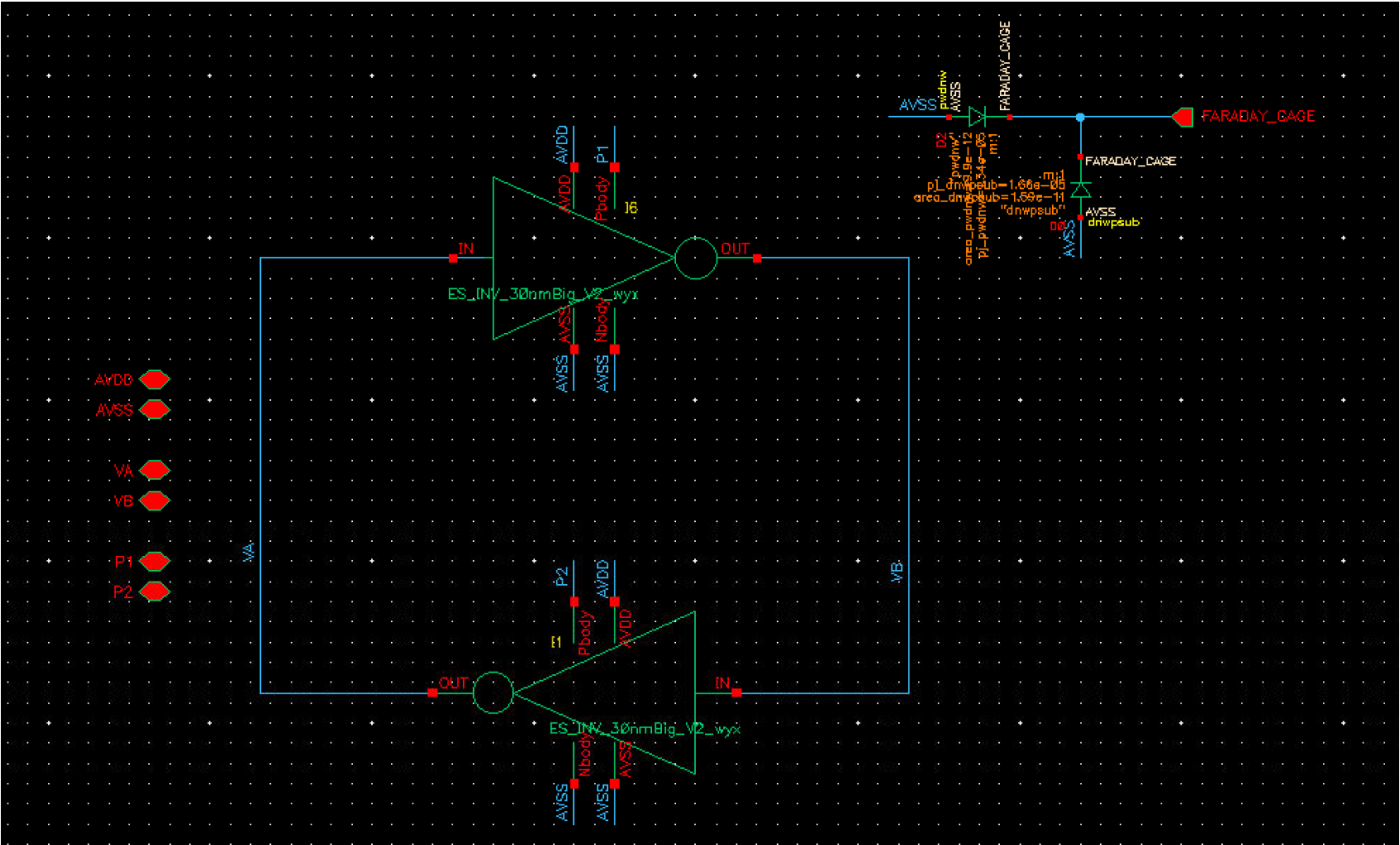
上图为MOM电容版图的参数变化；

1.14、版图举例-1 (tsmcN28)



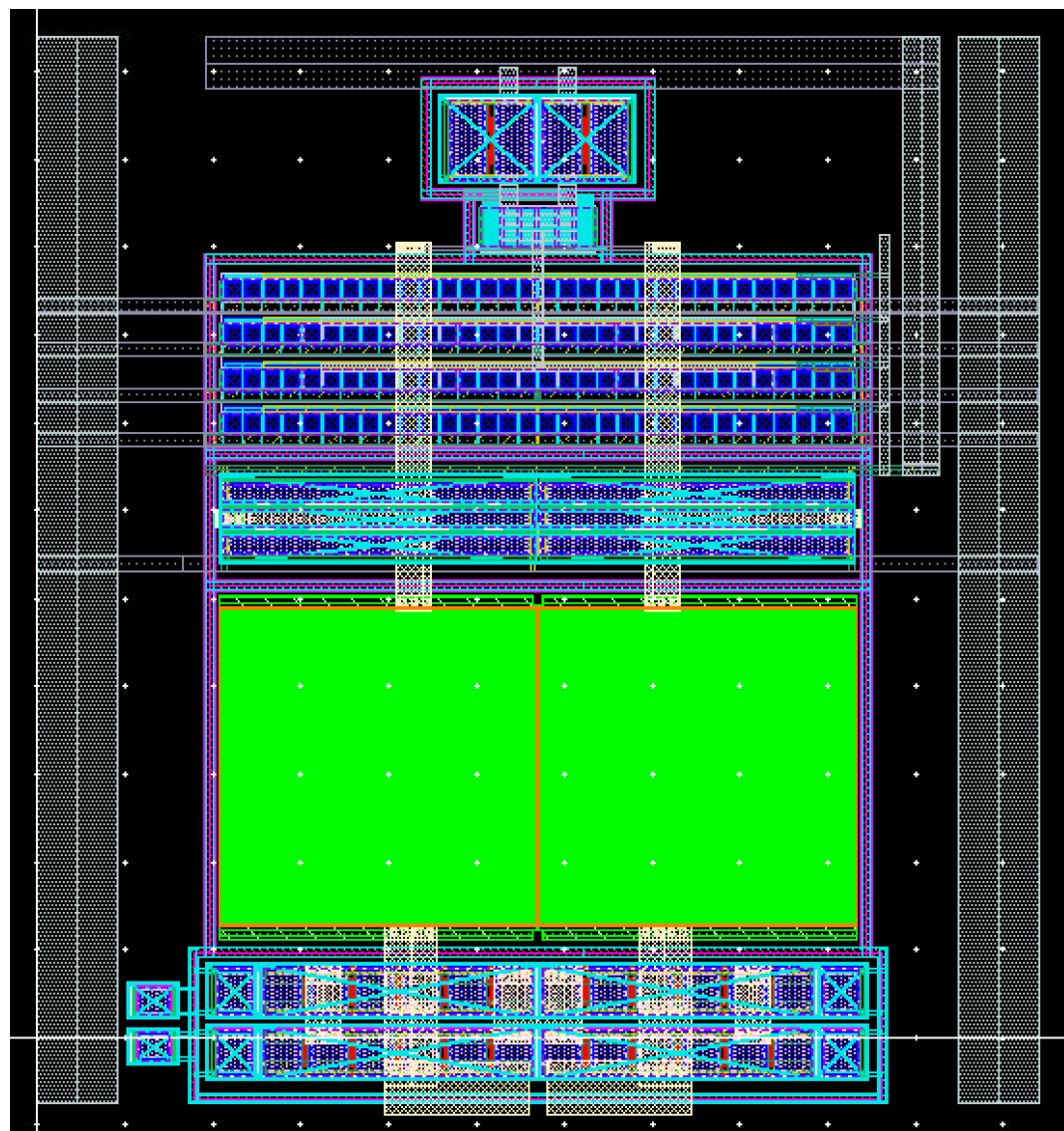
1、左图的版图大家可以思考一下电路是什么结构？

1.15、版图举例-1 (tsmcN28)



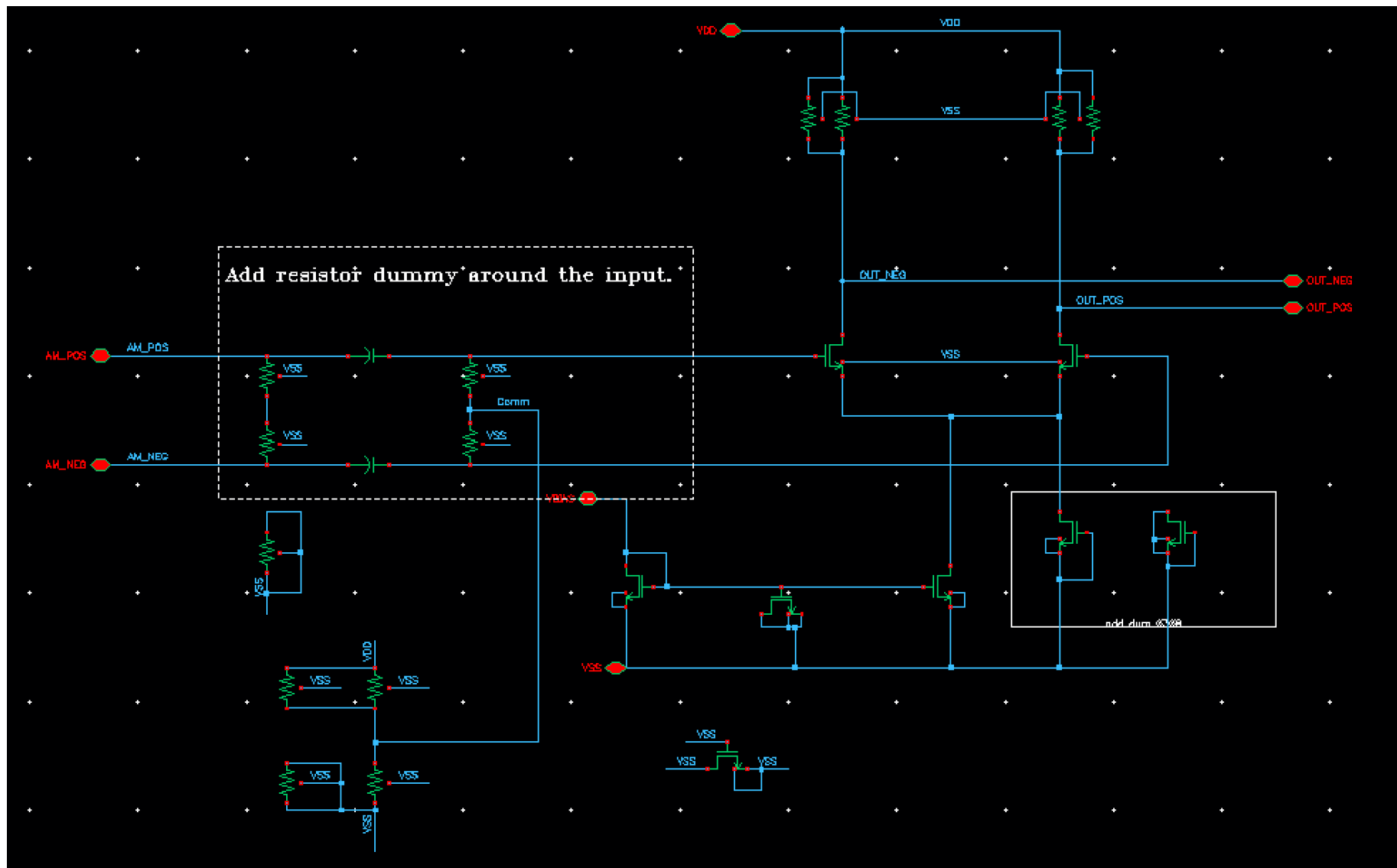
1、ES core;
Latch结构;

1.16、版图举例-2 (tsmcN28)



- 1、左图的版图为LVDS，目的是传输高速的差分信号；
- 2、版图较复杂，若不局部放大版图结构，较难区分；

1.17、版图举例-2 (tsmcN28)

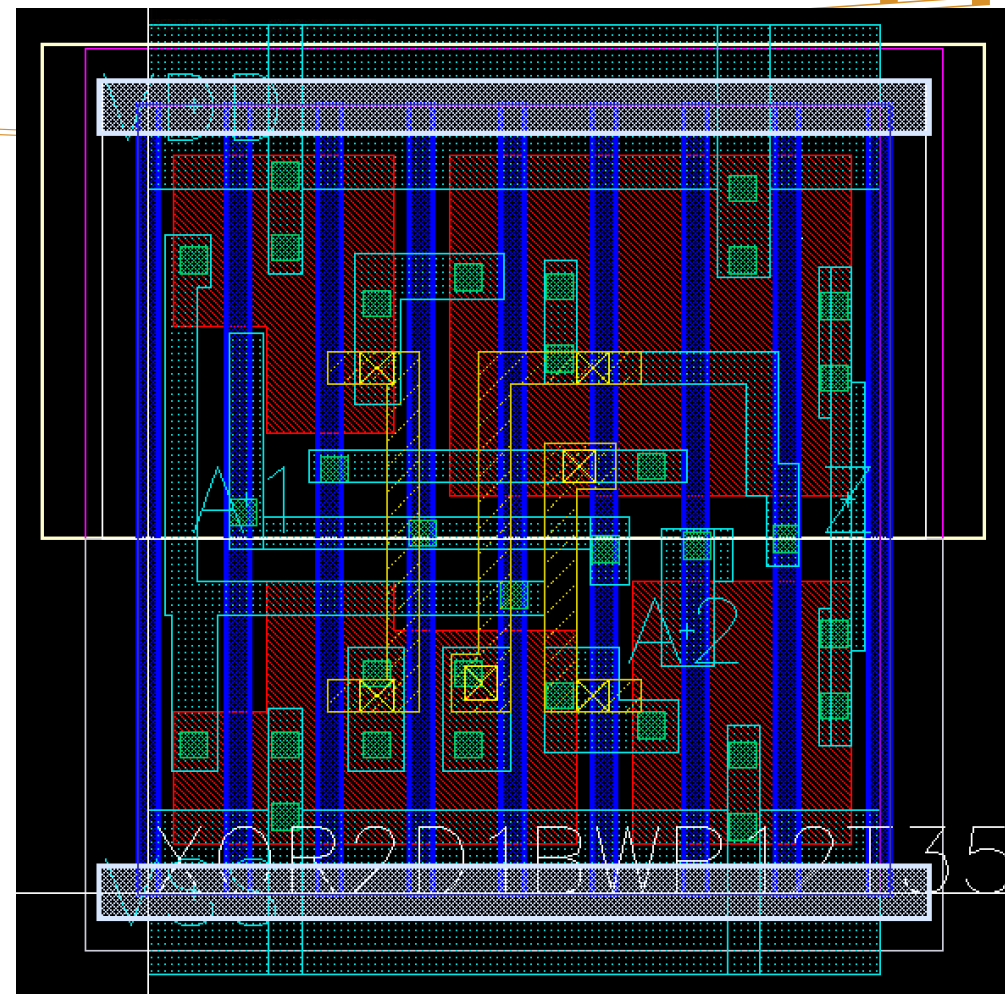
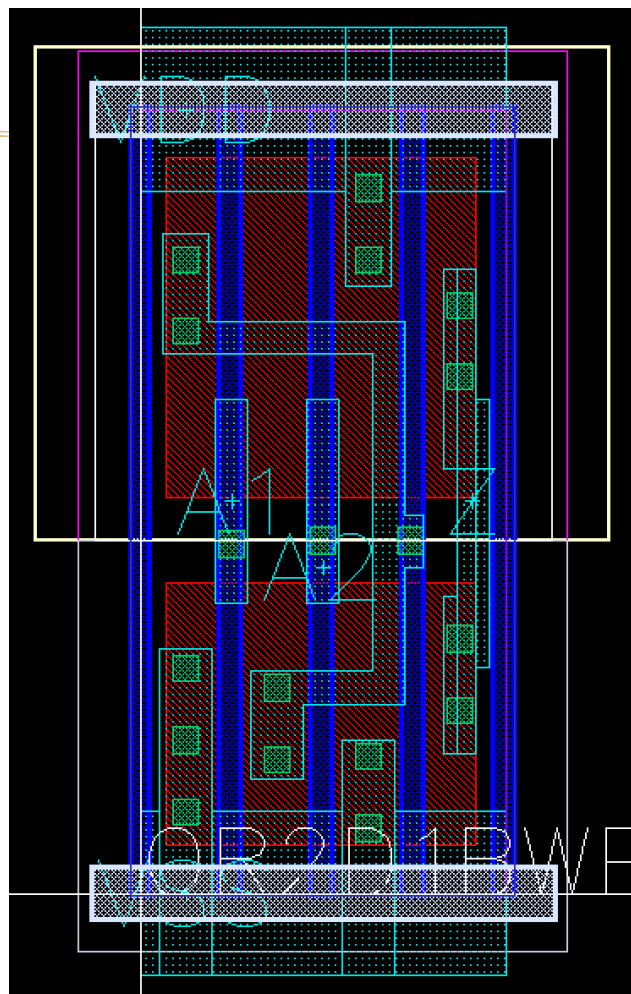
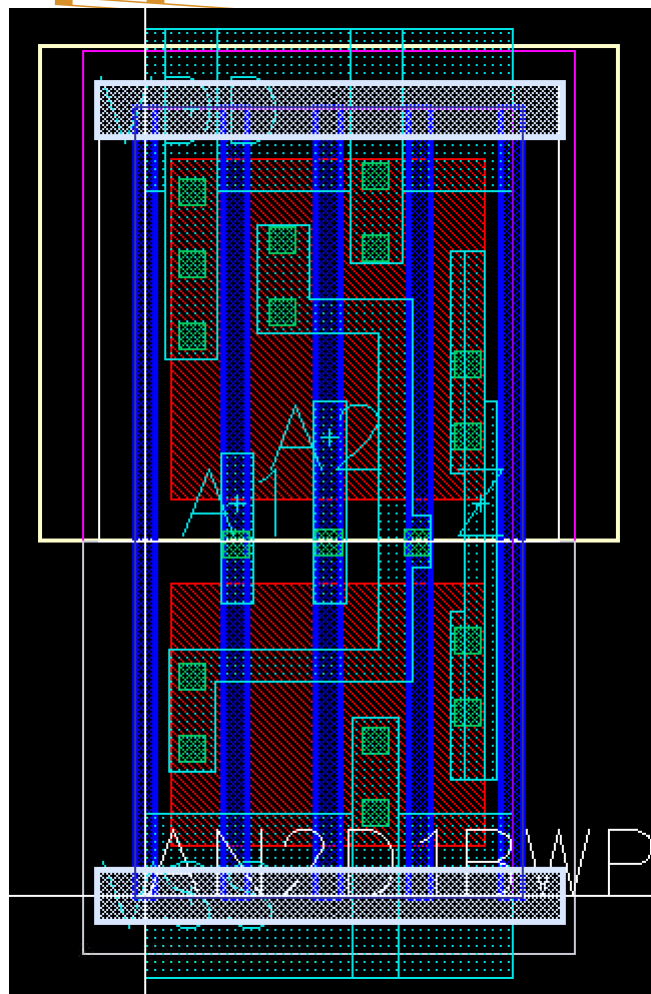


1、LVDS电路
如左图；

1.18、版图举例-3 (tsmcN28)



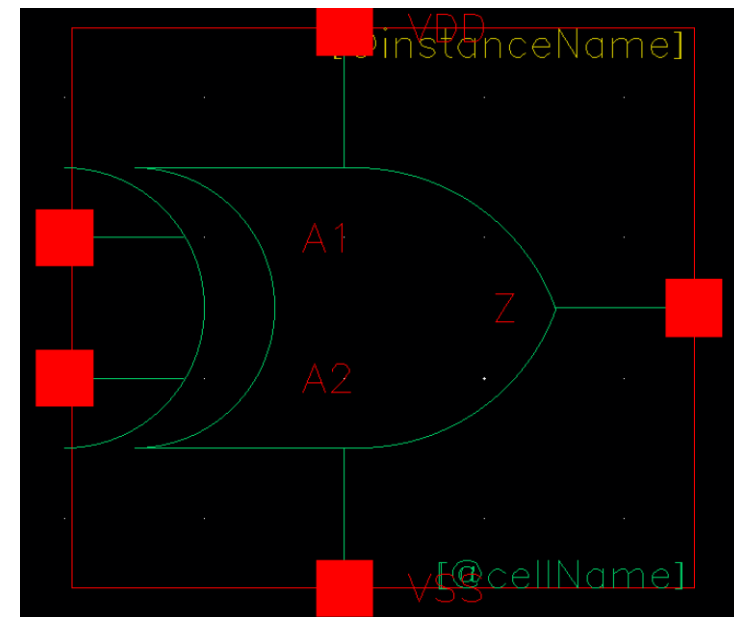
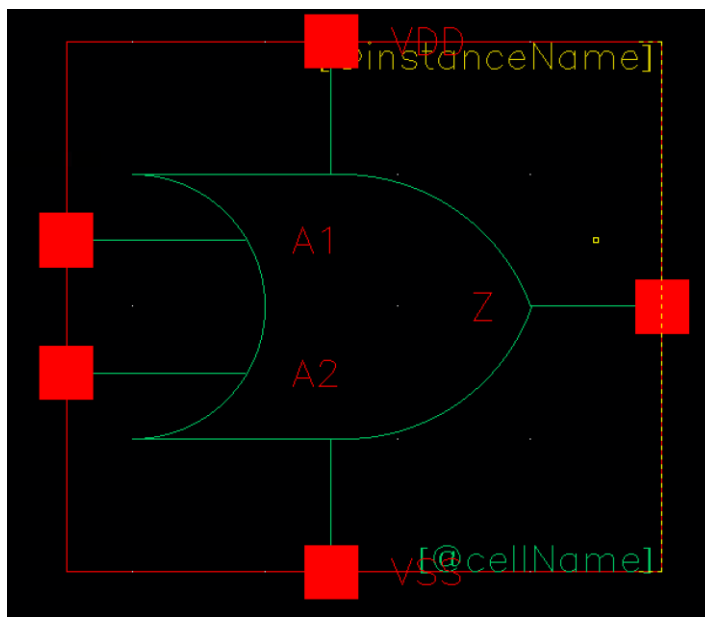
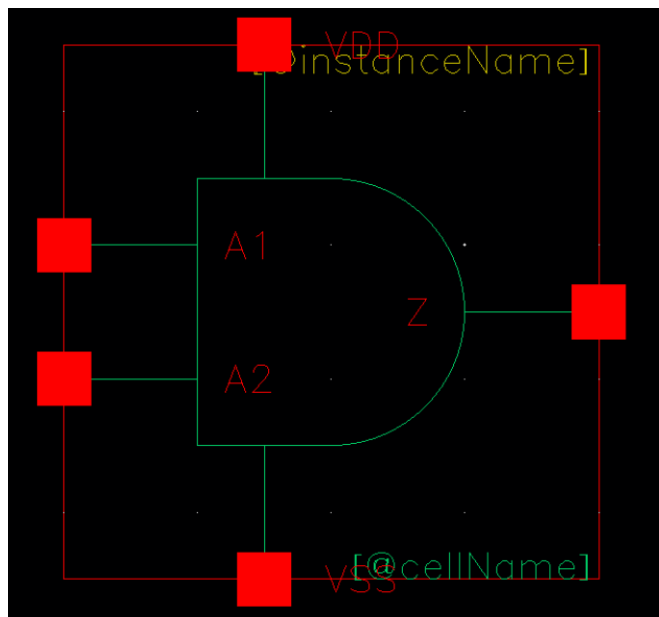
1、下图的版图大家可以思考一下电路是什么结构？



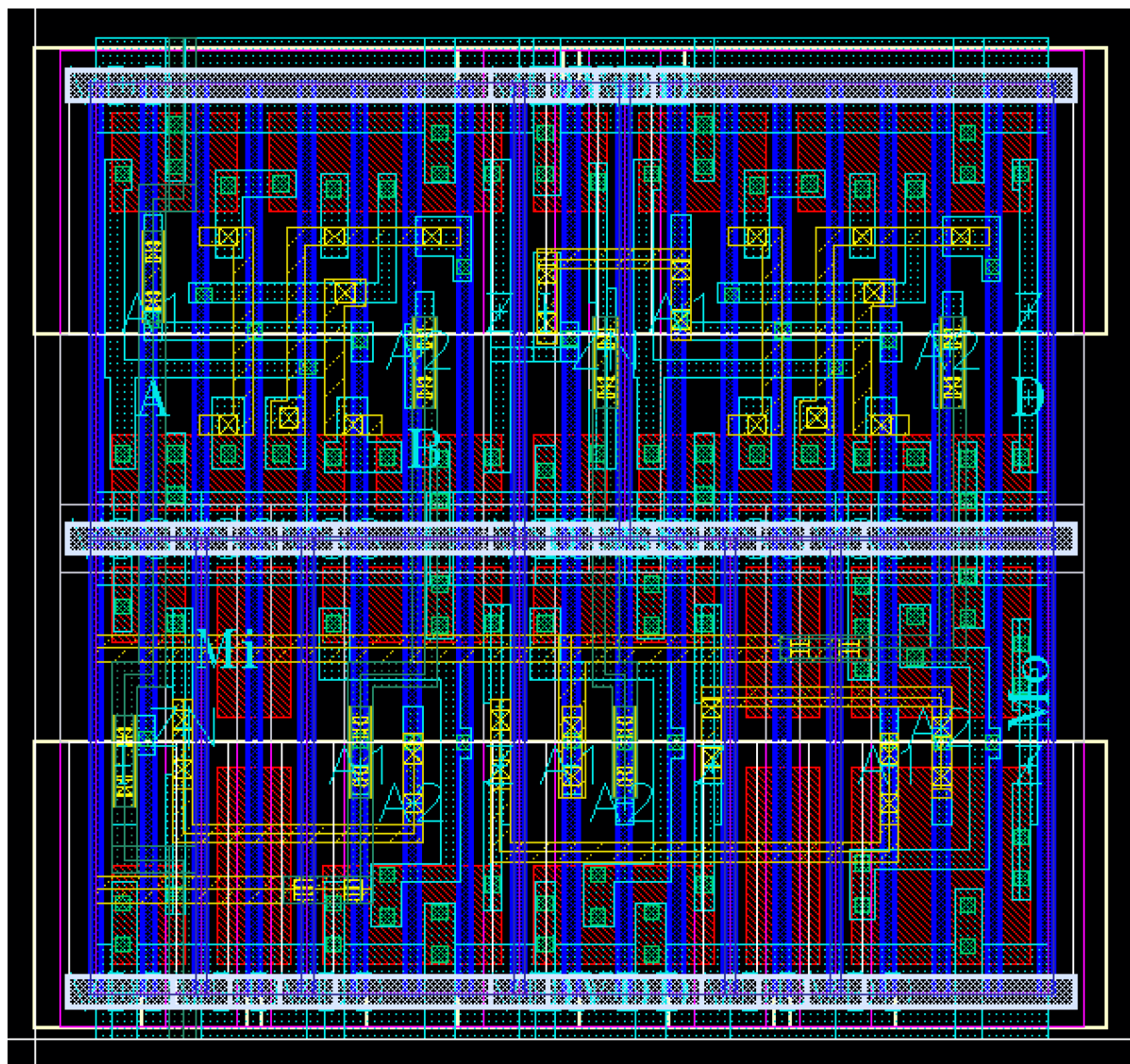
1.19、版图举例-3 (tsmcN28)



1、电路结构如下;



1.20、版图举例-4 (tsmcN28)

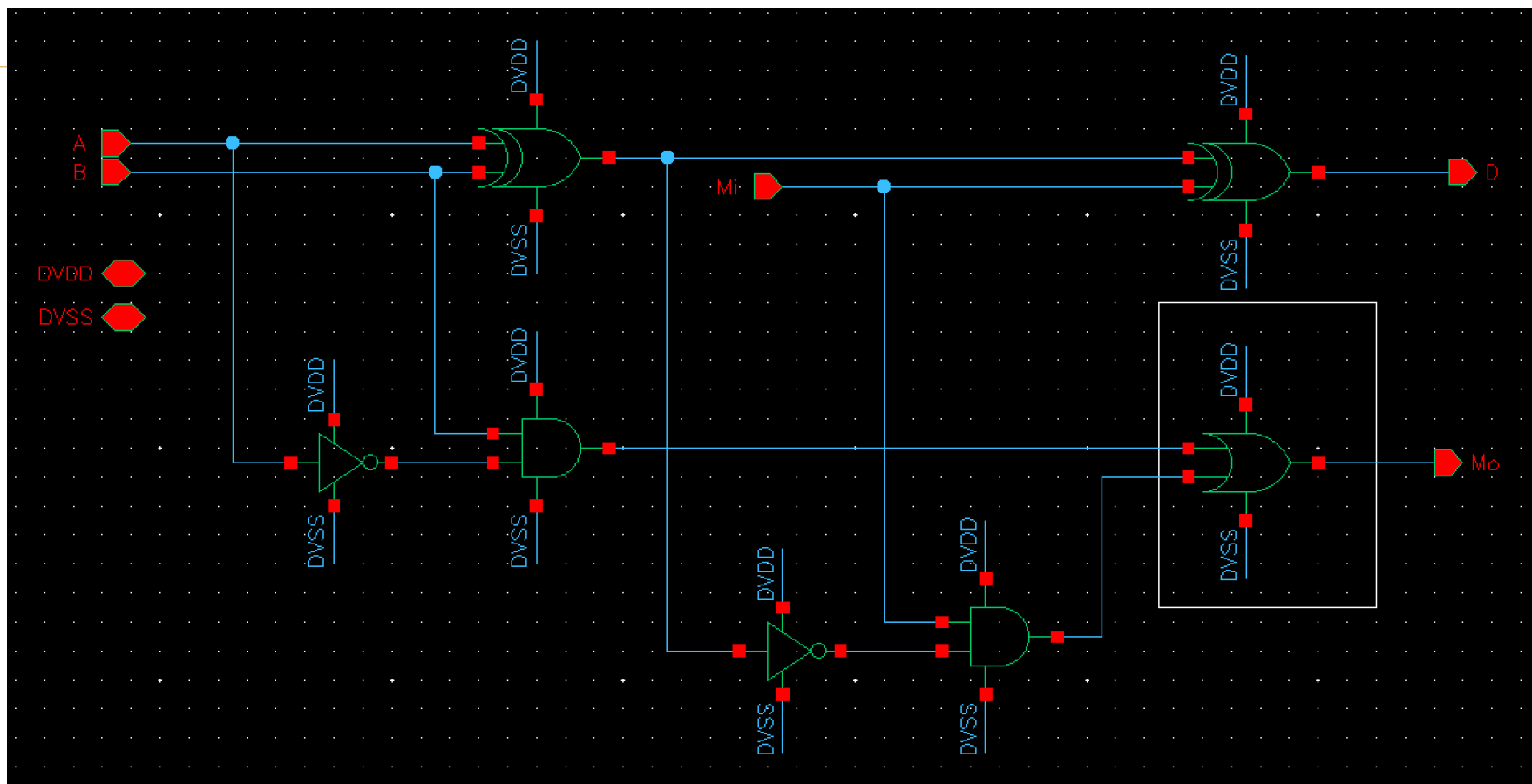


1、减法器版图;

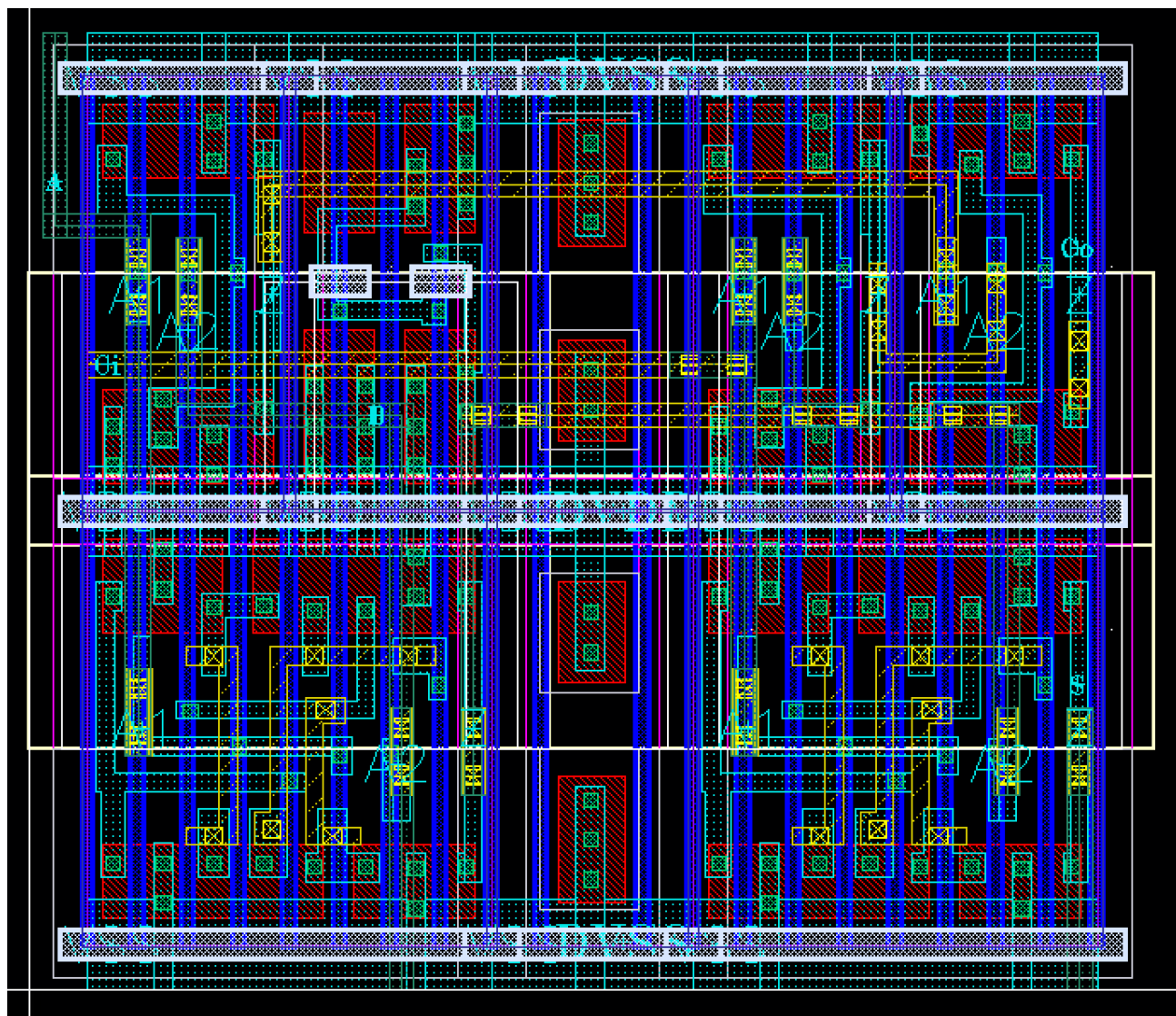
1.21、版图举例-4 (tsmcN28)



1、减法器电路；



1.22、版图举例-5 (tsmcN28)

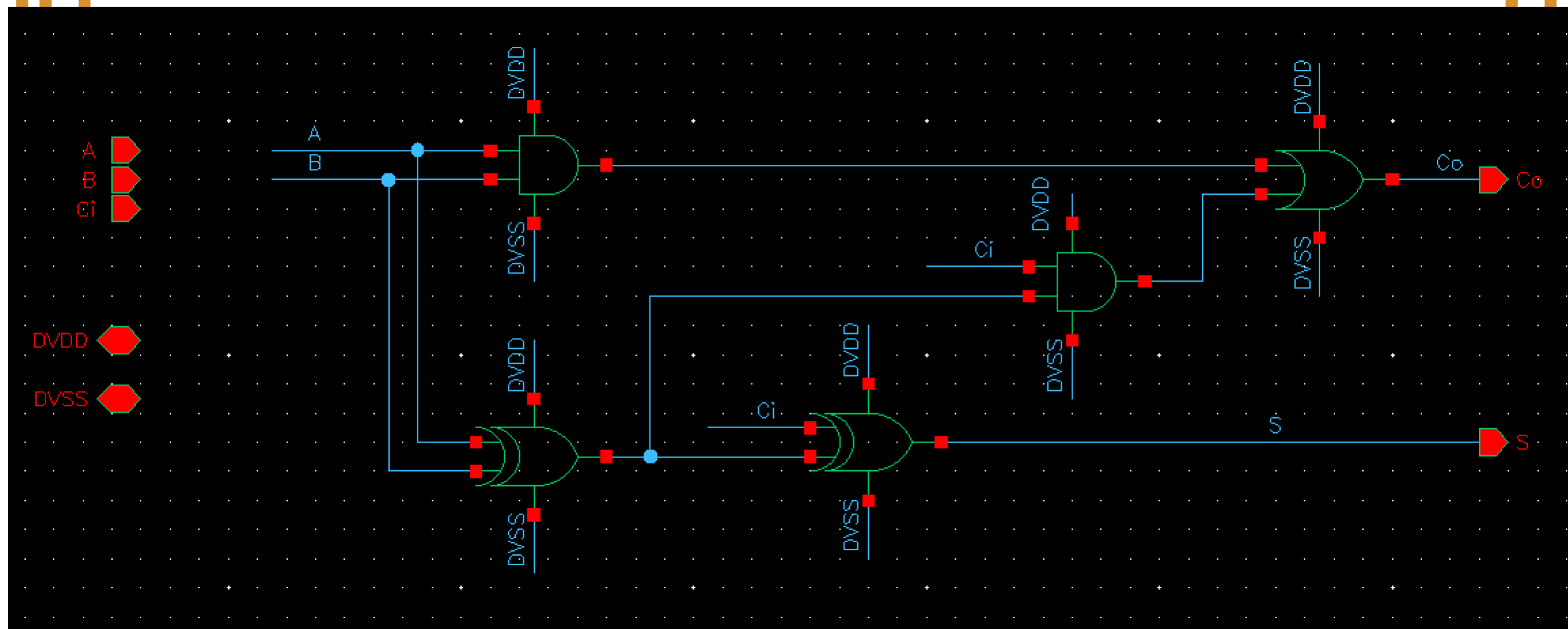


1、加法器版图;

1.23、版图举例-5 (tsmcN28)



1、加法器电路；





谢谢大家！